



Extrinsic Geometric Flows

Research Notes



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Introduction



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1. Heat Equation

The heat equation originated from Joseph Fourier's research on the theory of heat in 1807. He concluded that the heat change and heat transfer (heat flux $\mathbf{q}$) of a closed system must satisfy the following two laws:

- i. *The heat flux is proportional to the negative temperature gradient.*

$$\mathbf{q} = -\nabla u$$

- ii. *The heat energy $Q(\Omega, t)$ of a Euclidean domain $\Omega \subseteq \mathbb{R}^n$ at the time t can only be gained or lost via flux through its boundary.*

$$Q(\Omega, t_1) - Q(\Omega, t_2) = - \int_{t_1}^{t_2} \int_{\partial\Omega} \langle \mathbf{q}, \mathbf{n} \rangle \, dA dt$$

where $u = u(\mathbf{x}, t)$ is the temperature at the point $\mathbf{x} \in \Omega$. By these two laws we can get a partial differential equation as below, which is called the heat equation

$$\partial_t u - \Delta u = 0 \tag{1.1}$$

where $\Delta u = \operatorname{div}(\nabla u)$.

1.1 Gradient Flow

We say that a closed system reaches thermal equilibrium when there is no longer any change in heat, which corresponds to $\partial_t u = 0$, and that implies that

$$-\Delta u = 0$$

The solution of this equation is called the harmonic function. Now we define the Dirichlet Energy of the temperature function $u(\mathbf{x}, t)$ as $E(u) = (1/2) \int_{\Omega} |\nabla u|^2 d\mu$ it is a potential energy function. Then its stationary point is exactly the temperature function that satisfies thermal equilibrium conditions. Indeed, if $U \in C^\infty[\Omega \times (-\epsilon, \epsilon)]$ is a smooth variation of u with u outside a compact set, then, setting $\varphi(\mathbf{x}) = \partial_t|_{t=0} U$ we find that

$$\left. \frac{dE(U)}{dt} \right|_{t=0} = \int_{\Omega} \nabla \varphi \cdot \nabla u \, d\mu = - \int_{\Omega} \varphi \Delta u \, d\mu$$

This calculation also shows that the heat equation is a **Gradient Flow** of the Dirichlet Energy: As a system spontaneously evolves towards an equilibrium state, its energy function $E(u)$ will inevitably gradually minimize along the gradient direction, and the final point of this process is the equilibrium point, or the stationary point of $E(u)$.

Therefore, gradient flow is actually a trend, an evolutionary process of a system, in which the system's energy function $E(u)$ gradually decreases with time along the negative gradient direction. This flow can usually be described by a partial differential equation, and each solution is a characterization of a certain state of the system satisfying this evolutionary trend under different initial conditions (a function of time t). This leads to our rigorous definition of gradient flow.

Definition 1.1 (Gradient Flow). *Gradient flow is an evolutionary trend of a dynamic system. In this process, the system's energy function $E \in C^\infty$ gradually decreases with time along the negative gradient direction, which is characterized by the initial value problem of the following partial differential equation.*

$$\begin{cases} u'_t = -\nabla E(u), & t > 0 \\ u(\mathbf{x}, 0) = u_0(\mathbf{x}) \end{cases} \quad (1.2)$$

It's solution is called a gradient flow curve.

From this, we can extend the concept of geometric flow, which is also a mode of evolution that can be described by some partial differential equations. However, this mode of evolution operates on geometric objects (such as curves, surfaces, and manifolds) rather than dynamic systems. The energy function driving this evolution is certain geometric quantities (such as area, arc length of curves, and curvature). Under this trend, these geometric bodies evolve (deform) in the direction that minimizes these geometric quantities. Everything we will discuss thereafter revolves around this concept.

Geometric flows are further divided into external curvature flows and internal curvature flows. External curvature flows are characterized by a submanifold evolving in an ambient space, with velocity determined by its extrinsic curvature. This is the focus of our discussion. Internal curvature flows, on the other hand, are mathematical evolutions of a Riemannian metric (the shape of a manifold) that depend only on the internal geometry, not on how it is embedded in higher-dimensional space.

2. Hypersurfaces in Euclidean Space

Definition 2.1 (Hypersurface). Let M^n be an n -dimensional smooth manifold, N be an $(n+1)$ dimensional smooth manifold, and let

$$X : M^n \rightarrow N \quad (2.1)$$

be a smooth immersion (that is, a smooth map whose differential $dX : TM^n \rightarrow TN$ is everywhere injective). Then the pair (M^n, X) is defined as a hypersurface in N .

We say that a map X between two manifolds M^n and N is proper if its inverse X^{-1} satisfies that any compact subset $K \subset N$ its preimage $X^{-1}(K)$ is also a compact set in M^n .

2.1 Mean Curvature

Given a vector $\mathbf{u} \in \mathbb{R}^{n+1}$, the **directional derivative** in the direction $\mathbf{u}$ is denoted by $D_{\mathbf{u}}$. It is an operator such tht it acts on a smooth function $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ by

$$D_{\mathbf{u}} : f \mapsto u^i \frac{\partial f}{\partial x^i}$$

and on a smooth vector field $V : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ by

$$D_{\mathbf{u}} : V \mapsto u^i \frac{\partial V^j}{\partial x^i} \cdot \frac{\partial}{\partial x^j}$$

Since Euclidean space $\mathbb{R}^{n+1}$ is flat, the tangent space at any point possesses a natural isomorphism with the space itself (e.g, translation) $\sigma : T\mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$; the so-called Euclidean covariant derivative is defined as the result of any directional derivative under the action of this isomorphism, which we denote by $\widetilde{D}_{\mathbf{u}} = \sigma \circ D_{\mathbf{u}}$. If we also denote that $\tilde{\mathbf{u}} = \sigma(\mathbf{u})$ then it's easy to verify that for any smooth vector field $V : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ there are

$$\widetilde{D}_{\mathbf{u}}(V) = D_{\tilde{\mathbf{u}}}(\tilde{V})$$

Definition 2.2 (Connection). For a proper smooth immersion $X : M \rightarrow \mathbb{R}^{n+1}$ such that $X(M) = \mathcal{M}$ we define a connection on the smooth manifold M be a real bilinear map $\nabla : TM \times \Gamma(TM) \rightarrow TM$ which:

- respects the bundle structure, in the sense that $\pi(\nabla_{\mathbf{u}}V) = \pi(\mathbf{u})$ for any $\mathbf{u} \in TM$ and $\pi : TM \rightarrow M$ is the projection of the tangent bundle over M ,

- is smooth, in the sense that $\nabla_U V \in \Gamma(TM)$ for each pair $U, V \in \Gamma(TM)$ where

$$\nabla_U V(p) = \nabla_{U_p} V = \nabla(U_p, V)$$

- satisfies the Leibniz rule.

where $\Gamma(TM)$ denotes all the smooth vector fields on TM . For a fixed $\mathbf{u} \in TM$, the induced derivation $\nabla_{\mathbf{u}} : \Gamma(TM) \rightarrow TM$ is called the covariant derivative in the direction $\mathbf{u}$.

Obviously, the direction derivative is a Euclidean connection. Then we define the Euclidean pulback connection D on the pulback bundle $X^*T\mathbb{R}^{n+1}$ (such that its fiber on $p \in M$ is $T_{X(p)}\mathbb{R}^{n+1}$) as

$$D_{\mathbf{u}}(X^*V) = D_{dX(\mathbf{u})}(V)$$

for any tangent vector $\mathbf{u} \in TM$ and any vector field $V \in \Gamma(T\mathbb{R}^{n+1})$.

Observe that in the embedded case the restriction bundle $T\mathbb{R}^{n+1}|_{\mathcal{M}}$ splits as an orthogonal sum $TM \oplus^\perp NM$, where NM is the normal bundle. Then for any $\mathbf{u} \in T\mathbb{R}^{n+1}$ we define the normal and tangential projections as

$$\mathbf{u}^\perp = \mathbf{u} - \langle \mathbf{u}, \boldsymbol{\nu} \rangle \boldsymbol{\nu}, \quad \mathbf{u}^\top = \langle \mathbf{u}, \boldsymbol{\nu} \rangle \boldsymbol{\nu}$$

where $\boldsymbol{\nu}$ is either of the two unit vectors in $N_p\mathcal{M}$ and $p = \pi(u)$. Then pulling it back to M we have $X^*T\mathbb{R}^{n+1} \cong TM \oplus^\perp NM$.

Lemma 2.1. *If $\mathcal{M}^n \hookrightarrow \mathbb{R}^{n+1}$ is an embedded hypersurface, then the tangential projection of the Euclidean connection D*

$$\nabla_{\mathbf{u}} V = (D_{\mathbf{u}} V)^\top \tag{2.2}$$

defines a connection $\nabla : TM^n \times \Gamma(TM^n) \rightarrow TM^n$ on $\mathcal{M}^n$. It is metric compatible and symmetric.

Thus, by pulling it back to M we can get a connection $\nabla : TM^n \times \Gamma(TM^n) \rightarrow TM^n$ via

$$dX(\nabla_{\mathbf{u}} V) = (D_{\mathbf{u}}[dX(V)])^\top \tag{2.3}$$

where D is the Euclidean pulback connection on $X^*T\mathbb{R}^{n+1}$. It can be verify that the pulback connection ∇ is the Levi-Civita connection of the Riemannian metric g .

On a local coordinate $\{x^i\}$ we introduce the Christoffel symbols $\Gamma_{ij}^k : U \rightarrow \mathbb{R}$ to depict the connection, which is defined by

$$\Gamma_{ij}^k \partial_k = \nabla_i \partial_j \tag{2.4}$$

Since by (2.3) we have that $dX(\nabla_i \partial_j) = [D_i(\partial_j X)]^\top$ and also

$$dX(\nabla_i \partial_j) = dX(\Gamma_{ij}^k \partial_k) = \Gamma_{ij}^k \partial_k X$$

then for any $\partial_l X$ we have

$$\langle [D_i(\partial_j X)]^\top, \partial_l X \rangle = \langle D_i(\partial_j X), \partial_l X \rangle = \langle \partial_i \partial_j X, \partial_l X \rangle$$

On the other hand, by the Riemannian metric g we have that $\Gamma_{ij}^k \langle \partial_k, \partial_l \rangle = \Gamma_{ij}^k g_{kl}$. If we let $g^{kl} \cdot g_{kl} = 1$ then we have that

$$\Gamma_{ij}^k = g^{kl} \langle \partial_i \partial_j X, \partial_l X \rangle$$

Now we define the normal part of the Euclidean covariant derivative $(D_u V)^\perp$ along $\mathcal{M}$ as the second fundamental form of $\mathcal{M}$.

Lemma 2.2. *Let $u, v \in T\mathcal{M}$, and let $V \in \Gamma(T\mathcal{M})$ be an extension of v . Then the smooth map $\mathbf{\Pi} : T\mathcal{M} \times T\mathcal{M} \rightarrow N\mathcal{M}$ defined by*

$$\mathbf{\Pi}(u, v) = (D_u V)^\perp \quad (2.5)$$

is well-defined and bilinear. It also happens to be symmetric.

Pulling back to M we obtain the symmetric, bilinear second fundamental form $\mathbf{h} : TM \times TM \rightarrow NM$ of $X : M \rightarrow \mathbb{R}^{n+1}$

$$\mathbf{h}(u, v) = (D_u V)^\perp \quad (2.6)$$

where $V \in \Gamma(TM)$ is any extension of v .

Choosing a unit normal $\boldsymbol{\nu} \in N_x M = N_p \mathcal{M}$, where $p = X(x)$ we can write

$$\mathbf{h}_x(u, v) = -h_x(u, v) \cdot \boldsymbol{\nu} = -\Pi_p(u, v) \boldsymbol{\nu} = \mathbf{\Pi}_p(u, v)$$

Now given $u, v \in T_x M$, if we define the derivative of dX as $\nabla^2 X(u, v) = (\nabla_u dX)(v)$ then by the following Leibniz rule

$$\nabla_u [dX(V)] = (\nabla_u dX)(v) + dX(\nabla_u V)$$

we could have that

$$\nabla^2 X = D_u [dX(V)] - dX(\nabla_u V) = [D_u (dX(V))]^\perp$$

It follows that

$$\nabla^2 X = \mathbf{h}$$

or, in components with respect to local bases $\{e_i\}_{i=1}^n$ for $T_x M$ and $\boldsymbol{\nu}$ for $N_x M$,

$$\nabla_i \nabla_j X = -h_{ij} \boldsymbol{\nu}$$

To each symmetric bilinear form corresponds a selfadjoint endomorphism. The **Weingarten map** $L_p : T_p\mathcal{M} \rightarrow T_p\mathcal{M}$ of $\mathcal{M}$ (corresponding to the choice of unit normal $\boldsymbol{\nu}$ at p) is the linear map defined by

$$L_p(\mathbf{u}) = D_{\mathbf{u}}N$$

where $N \in \Gamma(N\mathcal{M})$ of $\boldsymbol{\nu}$ satisfying $|N| \equiv 1$ in a neighborhood of p . It is well-defined because

$$0 = \frac{1}{2}D_{\mathbf{u}}|N|^2 = \langle D_{\mathbf{u}}N, \boldsymbol{\nu} \rangle$$

and also since for any $v \in T_p\mathcal{M}$ we have that $\langle N, v \rangle = 0$, then by the Leibniz rule we have that

$$0 = \nabla_{\mathbf{u}} \langle N, v \rangle = \langle D_{\mathbf{u}}N, v \rangle + \langle \boldsymbol{\nu}, D_{\mathbf{u}}V \rangle$$

From (2.5) we have that

$$\langle D_{\mathbf{u}}N, v \rangle = -\langle \boldsymbol{\nu}, D_{\mathbf{u}}V \rangle = \mathbf{II}(u, v)$$

Pulling back the map L_p to M^n , we obtain the Weingarten tensor $L_x : T_xM \rightarrow T_xM$ of $X : M^n \rightarrow \mathbb{R}^{n+1}$,

$$L_x(\mathbf{u}) = D_{\mathbf{u}}N$$

where $N \in \Gamma(NM)$ of some $\boldsymbol{\nu}$ satisfying $|N| \equiv 1$ in a neighborhood of p . Let g be the Riemannian metric on M then we have that

$$\begin{aligned} \mathbf{II}(u, v) &= (D_{\mathbf{u}}V)^\perp = \langle D_{\mathbf{u}}N, v \rangle \\ &= \langle L_p(\mathbf{u}), v \rangle = g[L_x(\mathbf{u}), v] = \mathbf{h}_x(u, v) \end{aligned}$$

With respect to local coordinates $\{x^i\}_{i=1}^n$ on M^n , let $L_x(\partial_i) = L_i^k \partial_k X$ then we have

$$\mathbf{II}_{ij} := \mathbf{II}(\partial_i, \partial_j) = g\left(L_i^k \partial_k, \partial_j\right) = L_i^k g_{kj}$$

Thus, by $L_i^k = \mathbf{II}_{ij} g^{kj}$ there are

$$D_i N = L_x(\partial_i) = L_i^k \partial_k X = \mathbf{II}_{ij} g^{kj} \partial_k X \quad (2.7)$$

If $\mathcal{M}$ is orientable, then we can choose a global unit normal field $N \in \Gamma\mathcal{M}$ in which case the tensor fields $\mathbf{h} \in \Gamma(T^*\mathcal{M} \otimes T^*\mathcal{M})$, $L \in \Gamma(T^*\mathcal{M} \otimes T\mathcal{M})$, $\mathbf{II} \in \Gamma(T^*\mathcal{M} \otimes T^*\mathcal{M})$, and $L \in \Gamma(T^*\mathcal{M} \otimes T\mathcal{M})$ are all well-defined globally. We can identify such vector field $N : \mathcal{M} \rightarrow N\mathcal{M} \subseteq T\mathbb{R}^{n+1}$ by the following **Gaussian map**

$$G : \mathcal{M} \rightarrow S^n \subseteq \mathbb{R}^{n+1}$$

via the canonical identification $T_p\mathbb{R}^{n+1} \cong \mathbb{R}^{n+1}$ and S^n is an n -dimensional sphere. Then $G(p) = N_p$

and the tangent spaces $T_p\mathcal{M}$, $T_{G(p)}S^n$ coincide under the identification $T_p\mathbb{R}^{n+1} \cong \mathbb{R}^{n+1}$.

$$T_{G(p)}S^n = \{v \in T_{G(p)}\mathbb{R}^{n+1} \mid \langle v, N_{G(p)} \rangle = 0\}$$

We define A_p the **shape operator** of $\mathcal{M}$ as the following derivative of Gaussian map

$$A_p := dG|_p: T_p\mathcal{M} \rightarrow T_{G(p)}S^n$$

Since for any curve $\gamma(t)$ that go through p and have the tangent vector $\mathbf{u}$ at the point p we have that

$$dG_p(\mathbf{u}) = \left. \frac{d}{dt} G(\gamma(t)) \right|_{t=0} = \left. \frac{d}{dt} N_{\gamma(t)} \right|_{t=0} = D_{\mathbf{u}}N = L_p(\mathbf{u})$$

Then there are $A = L$.

Definition 2.3 (Principal Curvature). *The (ordered) eigenvalues*

$$\kappa_1 \leq \kappa_2 \leq \cdots \leq \kappa_n \quad (2.8)$$

*of the Weingarten tensor are called the **principal curvatures** of $X : M \rightarrow \mathbb{R}^{n+1}$. The corresponding unit eigenvectors are called **principal directions**. We refer to an orthonormal basis of principal directions at a point as a **principal basis** and a local or global orthonormal frame field (not necessarily continuous) of principal directions as a **principal frame field**.*

For a Cylindrical surface $x(u, v) = (\cos v, \sin v, u)$ where $u, v \in \mathbb{R}$ we can find its tangent vectors as $X_u = \partial_u x = (0, 0, 1)$ and $X_v = \partial_v x = (-\sin v, \cos v, 0)$. Then the metric tensor can be written as

$$G = [g_{ij}]_{2 \times 2} = \begin{pmatrix} \langle X_u, X_u \rangle & \langle X_u, X_v \rangle \\ \langle X_v, X_u \rangle & \langle X_v, X_v \rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

So its inverse G^{-1} is also an identity matrix. The unit normal vector $\boldsymbol{\nu}$ of this surface can be calculated as

$$\boldsymbol{\nu} = X_u \times X_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1 \\ -\sin v & \cos v & 0 \end{vmatrix} = (-\cos v, -\sin v, 0)$$

By (2.6) we know that the components of the second fundamental form $\mathbf{h}$ should satisfies that $h_{ij} = \langle \nabla_i X_j, \boldsymbol{\nu} \rangle$. Then there are

$$\mathbf{h} = \begin{pmatrix} \langle \nabla_u X_u, \boldsymbol{\nu} \rangle & \langle \nabla_u X_v, \boldsymbol{\nu} \rangle \\ \langle \nabla_v X_u, \boldsymbol{\nu} \rangle & \langle \nabla_v X_v, \boldsymbol{\nu} \rangle \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$$

By (2.7) we can find the matrix of the Weingarten map as $L = -h_{ij}g^{ij} = -h_{ij}$. Thus the principal curvature of $x(u, v)$ is $\kappa_1 = 0$ and $\kappa_2 = 1$.

Definition 2.4 (Mean Curvature). We define the trace of the Weingarten tensor as the **mean curvature** and denote it by H , that is

$$H := \text{tr}(L) = \kappa_1 + \kappa_2 + \cdots \kappa_n = \text{tr}(L) \quad (2.9)$$

The mean curvature measures the local variation of area under small perturbations of the hypersurface. It is equal to the average of the second fundamental form over the tangent sphere and can also be related to the scalar part of the divergence of dX with respect to the induced covariant derivative and metric on $T^*M \otimes X^*TM$. We shall call a hypersurface $X : M^n \rightarrow \mathbb{R}^{n+1}$ **mean convex** if it admits a unit normal field with respect to which its mean curvature is nonnegative.

Now for any vector field or tensor field V we define the divergence of it as the trace of its covariant differential, that is

$$\text{div}V := \text{tr}(u \mapsto \nabla_u V) \quad (2.10)$$

Then we can also have that

$$\text{div}(dX) = \text{tr}[\nabla(dX)] = -g^{ij}h_{ij}\nu = -H \cdot \nu = \mathbf{H}$$

If we define the Laplacian Δf of a smooth function f as $\Delta f = \text{div}(\text{grad } f)$ then it's easy to know that $\mathbf{H} = \text{div}(dX) = \Delta X$ where $\text{grad } f$, the gradient of f is the tangent vector field dual to the differential df of f .

Definition 2.5 (Gaussian Curvature). We define the **Gaussian Curvature** K as the determinant of Weingarten tensor that is

$$K = \det(L) = \kappa_1 \cdot \kappa_2 \cdots \kappa_n = \det(L) \quad (2.11)$$

More generally, the m -th mean Curvature H_m , defined for $1 \leq m \leq n$ is the m -th elementary symmetric polynomial in the principal curvatures:

$$H_m := \sum_{1 \leq i_1 < i_2 < \cdots < i_m \leq n} \kappa_{i_1} \cdots \kappa_{i_m}$$

The **Harmonic mean curvature** H_{-1} is defined by

$$H_{-1} := \frac{1}{\kappa_1^{-1} + \cdots + \kappa_n^{-1}} = \frac{H_n}{H_{n-1}}$$

Example 2.1 (Round Euclidean spheres). Denote by S_r^n the n -sphere of radius r centered at the origin in $\mathbb{R}^{n+1}$ and let $\mathbf{N}(x) = x/|x|$ denote its unit outward normal vector field. The

second fundamental form and mean curvature of S_r^n are given by

$$\mathbf{II} = \frac{1}{r} \cdot \mathbf{I}_{S_r^n}, \quad H = \frac{n}{r}$$

Indeed for any $V, W \in TS_r^n$, by (2.5) we should have that

$$\begin{aligned} \mathbf{II}(V, W) &= (D_V W)^\perp = \langle D_V \mathbf{N}, W \rangle \\ &= \left\langle \frac{1}{r} D_V x, W \right\rangle = \frac{1}{r} \langle V, W \rangle \\ &= \frac{1}{r} \cdot \mathbf{I}_{S_r^n}(V, W) \end{aligned}$$

The orthogonal basis of the tangent space TS_r^n is $\{e_1, e_2, \dots, e_n\}$ then we know that g_{ij} and g^{ij} are both identify matrix then $L = \mathbf{II}$. Also, we have that

$$\mathbf{II}_{ij} = \mathbf{II}(e_i, e_j) = \frac{1}{r} \langle e_i, e_j \rangle = \frac{1}{r} \cdot \delta_{ij}$$

Thus the mean curvature of S_r^n should be

$$H = \text{tr}(L) = \frac{1}{r} \cdot \delta_{ij} = \frac{n}{r}$$

2.2 The Area and Volume Functional

Let A be a subset of $\mathbb{R}^{n+1}$, let m be a nonnegative integer, and let ω_m denote the volume of the m -dimensional unit Euclidean ball. For a subset $B \subseteq \mathbb{R}^{n+1}$, let $\text{diam}(B)$ denote the diameter of B and $\text{rad}(B) = (1/2) \cdot \text{diam}(B)$. Given $\delta > 0$, define

$$\mathcal{H}_\delta^m(A) = \inf_{\mathcal{B}} \sum_{B_i \in \mathcal{B}} \omega_m \cdot \text{rad}^m(B_i)$$

where the infimum is taken over all countable covers $\mathcal{B} = \{B_i\}$ of A satisfying

$$\max_{B_i \in \mathcal{B}} \text{diam}(B_i) \leq \delta$$

Obviously, $\mathcal{H}_\delta^m(A)$ is nonincreasing of δ .

Definition 2.6 (Hausdorff Measure). For any subset $A \subseteq \mathbb{R}^{n+1}$ we define its m -dimensional outer Hausdorff measure as follows

$$\bar{\mathcal{H}}^m(A) = \sup_{\delta > 0} \mathcal{H}_\delta^m(A) \quad (2.12)$$

Particularly, if the limit $\lim_{\delta \rightarrow 0} \mathcal{H}_\delta^m(A)$ exists then we say A is m -dimensional Hausdorff measurable and define its m -dimensional Hausdorff measure as

$$\mathcal{H}^m(A) = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^m(A) = \bar{\mathcal{H}}^m(A)$$

2.2.1 Basic Concepts

Definition 2.7 (Area). The area of a compact subset K of an embedded hypersurface $\mathcal{M}^n \hookrightarrow \mathbb{R}^{n+1}$ is defined as the n -dimensional Hausdorff measure on $\mathbb{R}^{n+1}$ of K . That is

$$\text{Area}(K) := \mathcal{H}^n(K) \quad (2.13)$$

Since it can be proved that the Hausdorff measure $\mathcal{H}^n$ is a regular outer Borel measure of $\mathbb{R}^{n+1}$, then any compact set K is Hausdorff measurable and thus such concept is well-defined.

With the concept of Hausdorff measure we can introduce the concept of Riemannian measure of a Riemannian manifold (M^n, g) as follows, which can help us doing some specific calculation.

Definition 2.8 (Riemannian Measure). For a Riemannian manifold (M^n, g) we take a locally finite atlas $\{(U_i, \varphi_i)\}_{i \in \mathcal{I}}$ for M_n and subordinate partition of unity $\{\rho_i\}_{i \in \mathcal{I}}$. Then for any compact subset $K \subseteq M^n$ let χ_K be the indicator function of K and g_{φ_i} is the component matrix of g with respect to the chart φ_i . We define the Riemannian measure of K as

$$\mu(K) = \sum_{i \in \mathcal{I}} \int_{U_i} [(\rho_i \cdot \chi_K) \circ \varphi_i^{-1}] \cdot \sqrt{\det g_{\varphi_i}} \, d\mathcal{H}^n \quad (2.14)$$

Since compact set is always Hausdorff measurable and it can be proved that such value of the integral is invariant under any choice of the atlas and the partition of unity, then such measure is well-defined. Now with the Riemannian measure we could compute the area of a compact hypersurface as follows

Lemma 2.3. Let $X : M^n \rightarrow \mathbb{R}^{n+1}$ be a compact immersed hypersurface then we have

$$\text{Area}(M) = -\frac{1}{n} \int_M \langle X, \mathbf{H} \rangle \, d\mu \quad (2.15)$$

Proof. By the definition of the Laplacian Δ and the divergence operator we know that

$$\begin{aligned}\frac{1}{2}\Delta|X|^2 &= \frac{1}{2}\Delta\langle X, X \rangle \\ &= \frac{1}{2} \cdot \operatorname{div}(\langle dX, X \rangle + \langle X, dX \rangle) = \operatorname{div}\langle dX, X \rangle\end{aligned}$$

From (2.10) we know that the divergence should satisfy the Leibniz rule, that is

$$\operatorname{div}\langle dX, X \rangle = \operatorname{tr}\langle dX, dX \rangle + \langle \operatorname{div}(dX), X \rangle = |dX|^2 + \langle \operatorname{div}(dX), X \rangle$$

Since it's well known that $\operatorname{div}(dX) = \mathbf{H}$ and

$$|dX|^2 = g^{ij}\langle dX(\partial_i), dX(\partial_j) \rangle = g^{ij}g_{ij} = n$$

Then we get that

$$\frac{1}{2}\Delta|X|^2 = \operatorname{div}\langle dX, X \rangle = n + \langle \mathbf{H}, X \rangle$$

Now with the natural definition of the area with Riemannian measure $\operatorname{Area}(M) = \int_M d\mu$ we get that

$$\begin{aligned}\operatorname{Area}(M) &= \frac{1}{n} \int_M n \, d\mu \\ &= \frac{1}{n} \int_M \left[\frac{1}{2}\Delta|X|^2 - \langle X, \mathbf{H} \rangle \right] d\mu\end{aligned}$$

By the divergence theorem we have

$$\int_M \frac{1}{2}\Delta|X|^2 \, d\mu = \int_M \operatorname{div}\langle dX, X \rangle \, d\mu \leq \int_{\partial M} \langle dX, \mathbf{N} \rangle \, d\sigma = 0$$

Then we get (2.15). ■

2.2.2 First variation of Area and Volume

Example 2.2. For a surface $M \subseteq \mathbb{R}^3$ parameterized by $X : \Omega \rightarrow \mathbb{R}^3$ the first variation of its area functional A_M should be that

$$\delta A[X, t] = - \int_M H \cdot \phi \, dA \quad (2.16)$$

where $X_t = X + t\phi \cdot \mathbf{N}$, $dA = \sqrt{g} \, d\sigma$, $d\sigma = du \wedge dv$ and $g = \det(g_{ij})$.

Let $g_{ij} = \langle \partial_i X, \partial_j X \rangle$, then we have

$$\begin{aligned} g_{ij}^t &= \left\langle \frac{\partial X_t}{\partial u_i}, \frac{\partial X_t}{\partial u_j} \right\rangle \\ &= \left\langle \frac{\partial X}{\partial u_i} + t \cdot \left(\phi \frac{\partial \mathbf{N}}{\partial u_i} + \frac{\partial \phi}{\partial u_i} \mathbf{N} \right), \frac{\partial X}{\partial u_j} + t \cdot \left(\phi \frac{\partial \mathbf{N}}{\partial u_j} + \frac{\partial \phi}{\partial u_j} \mathbf{N} \right) \right\rangle \\ &= g_{ij} + t\phi \cdot \left(\left\langle \frac{\partial X}{\partial u_i}, \frac{\partial \mathbf{N}}{\partial u_j} \right\rangle + \left\langle \frac{\partial \mathbf{N}}{\partial u_i}, \frac{\partial X}{\partial u_j} \right\rangle \right) + o(t) \\ &= g_{ij} - 2(t\phi)h_{ij} + o(t) \end{aligned}$$

Thus we could compute $\left(dg_{ij}^t / dt \right)_{t=0}$ as

$$\left. \frac{dg_{ij}^t}{dt} \right|_{t=0} = \lim_{t \rightarrow 0} \frac{g_{ij}^t - g_{ij}}{t} = -2\phi \cdot h_{ij}$$

Then the derivative of $g_t = \det(g_{ij}^t)$ is

$$\begin{aligned} \left. \frac{dg_t}{dt} \right|_{t=0} &= \frac{\partial g_t}{\partial g_{ij}^t} \cdot \left. \frac{dg_{ij}^t}{dt} \right|_{t=0} \\ &= \left(\frac{\partial}{\partial g_{ij}} \sum_k g_{ik} G_{ik} \right) \cdot (-2\phi \cdot h_{ij}) \\ &= G_{ij} \cdot (-2\phi \cdot h_{ij}) = (g_{ij})_{ji}^* \cdot (-2\phi \cdot h_{ij}) = g \cdot g^{ij} (-2\phi \cdot h_{ij}) = -2g (H \cdot \phi) \end{aligned}$$

Finally, by the definition of the first variation we have

$$\delta A[X, t] = \left. \frac{d}{dt} A_{X_t} \right|_{t=0} = \int_\Omega \left. \frac{d}{dt} \sqrt{g_t} \right|_{t=0} d\sigma = - \int_\Omega H \cdot \phi \sqrt{g} d\sigma$$

and that is (2.16).

3. Mean Curvature Flow

The mean curvature flow is (after choosing an appropriate ‘gauge’) a parabolic equation, with the minimal surface equation as its static case and elliptic counterpart. Its linear analogue is the heat equation, with its elliptic counterpart being the Laplace equation.

Definition 3.1 (Mean Curvature Flow). *A smooth 1-parameter family of immersions $X : M^n \times I \rightarrow \mathbb{R}^{n+1}$ satisfies the mean curvature flow if*

$$\partial_t X(\mathbf{x}, t) = \mathbf{H}(\mathbf{x}, t), \quad \forall (\mathbf{x}, t) \in M^n \times [0, T] \quad (3.1)$$

where $\mathbf{H}$ denotes the mean curvature vector of the family.

We say that $X : M \times I \rightarrow \mathbb{R}^{n+1}$ is a solution of **reparametrized mean curvature flow** if

$$[\partial_t X(\mathbf{x}, t)]^\perp = \mathbf{H}(\mathbf{x}, t)$$

In this case, the vector field

$$V(\mathbf{x}, t) := [\partial_t X(\mathbf{x}, t)]^\perp - \partial_t X(\mathbf{x}, t)$$

is tangential. Defining the 1-parameter family of self-diffeomorphisms $\varphi_t : M \rightarrow M$ by

$$\begin{cases} \partial_t \varphi_t(\mathbf{x}) = dX_t^{-1}[V(\varphi_t(\mathbf{x}), t)], \\ \varphi_0 = \text{id}_M \end{cases} \quad (3.2)$$

and let $\hat{X}_t = X_t \circ \varphi_t$. Then we claim that $\hat{X}$ is a solution of the mean curvature flow (3.1). This is because

$$\partial_t \hat{X}(\mathbf{x}, t) = \left[\frac{\partial}{\partial \mathbf{x}} X_t, \frac{\partial}{\partial t} X_t \right] \cdot \begin{bmatrix} \partial_t \varphi_t(\mathbf{x}) \\ \partial_t t \end{bmatrix} = dX_t(\varphi_t) \cdot \partial_t \varphi_t(\mathbf{x}) + \partial_t X_t(\varphi_t)$$

Combined with (3.2) and the relationship between differentiation and the action of tangent vectors, we can obtain that

$$\begin{aligned} dX_t(\varphi_t) \cdot \partial_t \varphi_t(\mathbf{x}) &= dX_t[dX_t^{-1}[V(\varphi_t, t)]] \\ &= V(\varphi_t, t) = [\partial_t X(\varphi_t, t)]^\perp - \partial_t X(\varphi_t, t) \end{aligned}$$

Hence, by the condition of the reparametrized mean curvature flow we have that

$$\partial_t \hat{X}(\mathbf{x}, t) = [\partial_t X(\varphi_t, t)]^\perp = \mathbf{H}(\varphi_t, t) = \mathbf{H}_{\hat{X}}(\mathbf{x}, t)$$

A much easier solution of the MCF (Mean Curvature Flow) is the following shrinking round sphere. Denote by $S_r^n \subseteq \mathbb{R}^{n+1}$ the n -sphere of radius $r > 0$ centered at the origin. Set $\mathcal{M}_0 = S_{r_0}^n$ where $r_0 > 0$. We seek a solution of a family of n -sphere to mean curvature flow, starting at $\mathcal{M}_0$. Then let $\mathcal{M}_t = S_{r(t)}^n$, it can naturally parameterized by an embedding $X_t : S_1^n \hookrightarrow$ as

$$X_t : \mathbf{x} \mapsto r(t) \cdot \mathbf{x}$$

Then by Example 2.1 we could calculate its mean curvature and $\partial_t X(\mathbf{x}, t)$ as

$$\mathbf{H} = -H(\mathbf{x}, t) \cdot \mathbf{N}(\mathbf{x}, t) = -n \frac{\mathbf{x}}{r(t)}, \quad \partial_t X(\mathbf{x}, t) = \frac{dr(t)}{dt} \mathbf{x}$$

Then the MCF (3.1) relates to an ODE

$$\frac{dr(t)}{dt} = -\frac{n}{r(t)}$$

One can solve this equation with initial value $r(0) = r_0$ as

$$r(t) = \sqrt{r_0^2 - 2nt}$$

Then we have

$$X_t(\mathbf{x}) = \mathbf{x} \cdot \sqrt{r_0^2 - 2nt}$$

Which means that the sphere in such family would shrink to a point for $t \in (-\infty, r_0^2/2n)$

Recall that a minimal hypersurface $\mathcal{M}^n \subseteq \mathbb{R}^{n+1}$ is a hypersurface satisfying $H \equiv 0$. Given a minimal surface $\mathcal{M}_0^n$ its clear that $\mathcal{M}_t \equiv \mathcal{M}_0$ is a solution of the mean curvature flow. In the following section we will discuss it specifically.

We claim that if $\mathcal{M}_t^{n-m}$ is a solution to mean curvature flow in $\mathbb{R}^{n-m+1}$, then the family of product hypersurfaces

$$\mathcal{M}_t^n := \mathcal{M}_t^{n-m} \times \mathbb{R}^m \subseteq \mathbb{R}^{n+1}$$

should be a solution to mean curvature flow in $\mathbb{R}^{n+1}$ which is called the cylinder over a solution. To see this let the immersion over $\mathcal{M}_t^{n-m}$ be $X'(\mathbf{x}, t)$ which satisfies that

$$\partial_t X'(\mathbf{x}, t) = \mathbf{H}_{n-m}$$

Then let the immersion $X : \mathcal{M}_t^n \rightarrow \mathbb{R}^{n+1}$ be defined as $X(\mathbf{x}, \mathbf{y}, t) = (X'(\mathbf{x}, t), \mathbf{y})$. Let $\partial/\partial x_i, i = 1, 2, \dots, n-m$ denote the linearly independent tangent vectors of $\mathcal{M}_t^{n-m}$ and $\partial/\partial y_j, j = 1, 2, \dots, m$ denote the tangent vector in $\mathbb{R}^m$. Thus we have

$$g_{ij}^X = \begin{bmatrix} g_{i_1 j_1}^{X'}(\mathbf{x}, t) & 0 \\ 0 & \delta_{i_2 j_2} \end{bmatrix}$$

where $\delta_{i_2 j_2} = 1$ if and only if $i_2 = j_2$ and $g_{i_1 j_1}^{X'}(\mathbf{x}, t)$ is the first fundamental form of $\mathcal{M}_t^{n-m}$ over X' . With this its easy to verify that the second fundamental form of $\mathcal{M}_t^n$ should be

$$\mathbf{II} = h_{ij}^X = \begin{bmatrix} h_{i_1 j_1}^{X'}(\mathbf{x}, t) & 0 \\ 0 & 0 \end{bmatrix}$$

Hence we can get that

$$\mathbf{H}_n = -g_X^{ij} \cdot h_{ij}^X = (\mathbf{H}_{n-m}, 0)$$

Therefore, we deduce that

$$\frac{\partial}{\partial t} X(\mathbf{x}, \mathbf{y}, t) = (\partial_t X'(\mathbf{x}, t), 0) = (\mathbf{H}_{n-m}, 0) = \mathbf{H}_n$$

3.1 Minimal Surface Problem

Now, let us briefly review the origins of the concept of mean curvature flow. First and foremost, let's consider a natural problem as follows, which is called **Minimal Surface Problem**. In $\mathbb{R}^3$, it can be expressed as:

Problem 3.1 (Plateau). Given a closed curve $\Gamma \in \mathbb{R}^3$, find a surface M with boundary Γ such that its area is minimal among all surfaces bounded by Γ .

To find such surface M let it be parameterized by $X : \Omega \rightarrow \mathbb{R}^3$ where the parameters $(u, v) \subseteq \Omega$ and $X|_{\partial\Omega} = \Gamma$ then its area can be written as

$$A_M = \int_{\Omega} |X_u \times X_v| \, du \, dv = \int_{\Omega} \sqrt{EG - F^2} \, d\sigma$$

where $d\sigma = du \wedge dv$ and $E = \langle X_u, X_u \rangle$, $F = \langle X_u, X_v \rangle$, $G = \langle X_v, X_v \rangle$. Then consider a deformation $X_t = X(u, v, t) = X + t\phi \cdot \mathbf{N}$ where ϕ is a compact support function $\phi : \Omega \rightarrow \mathbb{R}$ such that $\phi|_{\partial\Omega} = 0$ and $\mathbf{N}$ is the unit normal vector field of M . Then we know the first order variation of it should be

$$\delta(A_{X_t}) = \left. \frac{d}{dt} A_{X_t} \right|_{t=0} = - \int_{\Omega} H \cdot \phi \sqrt{g} \, d\sigma = - \int_M H \cdot \phi \, dA$$

Under the inner product $\langle f, g \rangle = \int_M f \cdot g \, dA$ in $L^2(M)$ we know that

$$\langle -H, \phi \rangle = \delta(A_{X_t}) = \langle \nabla A, \phi \rangle$$

Which shows $\nabla A = -H$. Thus the mean curvature flow of any $X : \Omega \times I \rightarrow \mathbb{R}^3$

$$\begin{cases} \partial_t X(u, v, t) = \mathbf{H} = H \cdot \mathbf{N} = -\nabla A \\ X|_{\partial\Omega}(u, v, t) = \Gamma, \end{cases} \quad \forall (u, v) \in \Omega, t \in I$$

represent a family of surfaces bounded by Γ such that their area decreased. Since M is the minimal surface, then by Fermat lemma its area variation should always be zero, that is

$$\delta(A_{X_t}) = \langle \nabla A, \phi \rangle \equiv 0, \quad \forall \phi$$

It implies that $\nabla A = -H \cdot \mathbf{N} \equiv 0$, then $H \equiv 0$. Thus, each minimal surface should necessarily be a stable point of a mean curvature flow. Particularly, in $\mathbb{R}^3$ we have that

$$H = \text{tr}(L) = \text{tr}(g^{ij} \cdot h_{ij}) = \frac{EN - 2FM + GL}{EG - F^2}$$

where $L = \langle X_{uu}, \mathbf{N} \rangle$, $M = \langle X_{uv}, \mathbf{N} \rangle$, $N = \langle X_{vv}, \mathbf{N} \rangle$. Then by $H = 0$ we can deduce the equation of a minimal surface $z = f(x, y)$ with the boundary $\Gamma : I \rightarrow \mathbb{R}^3$ should be

$$\begin{cases} \left[1 + (f'_y)^2\right] f''_x - 2f'_x f'_y f''_{xy} + \left[1 + (f'_x)^2\right] f''_{yy} = 0 \\ \Gamma_z(t) = f[\Gamma_x(t), \Gamma_y(t)] \end{cases} \quad (3.3)$$

Example 3.1. The plane and the following two surfaces are the classical solution of the Plateau's problem.

- (i) For any closed simple curve Γ on the following Catenoid, the Catenoid itself should be a minimal surface of such Plateau's problem with boundary Γ .

$$X(u, v) = (u, \cosh u \cos v, \cosh u \sin v) \quad (3.4)$$

- (ii) For any closed simple curve Γ on the following Helicoid, it is a minimal surface of such Plateau's problem with boundary Γ .

$$X(u, v) = (v \cos u, v \sin u, u) \quad (3.5)$$

One can verify this example by simply checking the mean curvature H of the surface (3.4) and (3.5) are both zero.

Furthermore, based on the concept of hypersurface area established in the previous section, we can generalize Plateau's problem to arbitrary n -dimensional hypersurfaces and demonstrate that a necessary condition for the solutions to these problems is that they constitute a stationary point of the mean curvature flow (we shall prove this by verifying that the mean curvature flow is the gradient flow of the area functional).

Problem 3.2 (Minimal Surface). For a smooth, compact and closed $(n - 1)$ -dimensional submanifold $\Gamma \subseteq \mathbb{R}^{n+1}$ find an n -dimensional hypersurface M which could be bounded by Γ and has the minimal area among all the hypersurface with the same boundary.

3.2 Self-similar Solutions

4. Isoperimetric Problem

4.1 The Case of Planar Curve

Problem 4.1 (Isoperimetric Problem). Given an area $A > 0$ in the plane $\mathbb{R}^2$, does there exist a closed curve with the shortest possible parameter such that the area enclosed within it is exactly A ?

We claim that such closed curve does exist, because the area enclosed by a closed curve of zero length is evidently also zero, it follows that the length of any closed curve enclosing an area $A > 0$ must necessarily be greater than zero. Furthermore, as it can be proven that the set comprising all closed curves with an enclosed area of A is complete, the Extreme Value Theorem implies that the lengths of all curves within this set must possess a greatest lower bound.

To find such minimal parameter, we are going to prove the following theorem, and thus the curve should be a circle and the minimal parameter is $L = 2\sqrt{\pi A}$.

Theorem 4.1. *For any closed planar curve α , its length L and enclosed area A should satisfy that*

$$L^2 \geq 4\pi A \quad (4.1)$$

and the equality holds if and only if α is a circle.

To prove this theorem, we plan to use the corresponding techniques of the curve shortening flow, which is a special case of the mean curvature flow. Specifically, we will first analyze the evolution of the following quantity along the curve shortening flow:

$$Q_\alpha(t) = L^2(t) - 4\pi \cdot A(t)$$

and conclude that it is monotonically non-increasing. Then we will show that regardless of the initial closed curve, curve shortening flow eventually evolves into some round points. Then since it's easy to verify that $Q_\alpha(t)$ is zero when α is a circle, we can conclude that $Q_\alpha(t) \geq 0$ for any closed curve α , and that is exactly (4.1).

As for the uniqueness of the equality condition, it is guaranteed by the evolutionary trend of the curve-shortening flow. Specifically, we will first demonstrate that any curve α satisfying $Q_\alpha = 0$ remains invariant under the curve-shortening flow. Subsequently—given that any curve undergoing curve-shortening flow eventually evolves into a round point—it follows that such α must be a circle.

4.1.1 The Evolution and Variation of Geometric Quantities

For a planar closed curve α let $X : M^1 \rightarrow \mathbb{R}^2$ be its smooth immersion parameterized by arc length s , where M^1 is isomorphic to the torus $\mathbb{S}^1 = \mathbb{R}/2\pi\mathbb{Z}$. Let $\mathbf{T}(s) = X'(s)$ and $\mathbf{N}(s) = -J \cdot \mathbf{T}$

be the unit tangent and normal vector fields along α respectively, where $J : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the counterclockwise rotation by $\pi/2$. Then let $\kappa(s)$ be the curvature of α at s , which is defined by the following Frenet-Serret equations:

$$\begin{cases} \mathbf{T}'(s) = -\kappa \cdot \mathbf{N}(s) \\ \mathbf{N}'(s) = \kappa \cdot \mathbf{T}(s) \end{cases} \quad (4.2)$$

Now let $\sigma : M^1 \rightarrow \mathbb{R}$ be the support function of α which is defined by:

$$\sigma : s \mapsto \langle X(s), \mathbf{N}(s) \rangle$$

Then by (4.2) we can get that

$$\int_{M^1} \sigma \kappa \, ds = \int_{M^1} \langle X, \kappa \cdot \mathbf{N} \rangle \, ds = - \int_{M^1} \langle X, X'' \rangle \, ds = \int_{M^1} |X'|^2 \, ds = L$$

If α is also embedded, then let $\mathcal{M} = X(M^1)$ and we write X as $X(s) = (x(s), y(s))$. By the Green's theorem and the fact that $\mathbf{T} = (x'(s), y'(s))$, $\mathbf{N} = (y'(s), -x'(s))$ we can get that

$$A = \frac{1}{2} \int_{\mathcal{M}} (x \, dy - y \, dx) = \frac{1}{2} \int_{M^1} (x, y) \cdot (y', -x') \, ds = \frac{1}{2} \int_{M^1} \langle X, \mathbf{N} \rangle \, ds = \frac{1}{2} \int_{M^1} \sigma \, ds$$

Lemma 4.1. *Let X be a curve evolving by*

$$\frac{\partial X}{\partial t} X(u, t) = -\phi(u, t) \cdot \mathbf{N}(u, t) \quad (4.3)$$

where $\phi : M^1 \times [0, T) \rightarrow \mathbb{R}$ is a function. Then for any C^2 function $f : M^1 \times [0, T) \rightarrow \mathbb{R}$ we have

$$f_{st} = f_{ts} + \phi \cdot f_s \cdot \kappa \quad (4.4)$$

Proof. By using

$$\frac{\partial}{\partial s} = \frac{\partial u}{\partial s} \cdot \frac{\partial}{\partial u} = \frac{1}{|X_u(u, t)|} \cdot \frac{\partial}{\partial u}$$

we can get that

$$\begin{aligned} f_{st} &= \frac{\partial}{\partial t} \left(\frac{1}{|X_u|} \cdot \frac{\partial f}{\partial u} \right) \\ &= -\frac{1}{|X_u|^3} \cdot \langle X_{tu}, X_u \rangle \cdot f_u + \frac{1}{|X_u|} f_{tu} \\ &= \frac{1}{|X_u|^3} [\phi_u \langle \mathbf{N}, X_u \rangle + \phi \langle \mathbf{N}_u, X_u \rangle] \cdot f_u + f_{ts} \\ &= \phi \cdot \langle \mathbf{N}_s, \mathbf{T} \rangle f_s + f_{ts} = \phi \cdot f_s \cdot \kappa + f_{ts} \end{aligned}$$

■

Now let $X : M^1 \times [0, T) \rightarrow \mathbb{R}^2$ be a smooth map, which is an immersion for each fixed $t \in [0, T)$. We say that $X(t)$ is a solution of the curve shortening flow if it satisfies that

$$\frac{\partial X}{\partial t}(u, t) = -\kappa(u, t) \cdot \mathbf{N}(u, t) \quad (4.5)$$

For such $X(u, t)$ we will prove the following lemmas, which describe the evolution of the length $L(t)$ and area $A(t)$ along the curve shortening flow.

Lemma 4.2. *Under the curve shortening flow, the unit tangent vector field $\mathbf{T}$ and the unit normal vector field $\mathbf{N}$ evolves by the following equations:*

$$\begin{cases} \mathbf{T}_t = -\kappa_s \cdot \mathbf{N} \\ \mathbf{N}_t = \kappa_s \cdot \mathbf{T} \end{cases} \quad (4.6)$$

Moreover, the curvature κ evolves by the following equation:

$$\kappa_t = \kappa_{ss} + \kappa^3 \quad (4.7)$$

Proof. By Lem.4.1 we know that under the curve shortening flow (4.5) there are

$$X_{st} = X_{ts} + \kappa^2 \cdot X_s$$

Then from the Frenet-Serret equations we can get that

$$\begin{aligned} \mathbf{T}_t &= (-\kappa \cdot \mathbf{N})_s + \kappa^2 \mathbf{T} \\ &= -\kappa_s \cdot \mathbf{N} - \kappa \cdot \mathbf{N}_s + \kappa^2 \mathbf{T} \\ &= -\kappa_s \cdot \mathbf{N} \end{aligned}$$

Since $\mathbf{N}_s = \kappa \cdot \mathbf{T}$, by

$$\frac{d}{dt} \langle \mathbf{N}, \mathbf{T} \rangle = \langle \mathbf{N}_t, \mathbf{T} \rangle + \langle \mathbf{N}, \mathbf{T}_t \rangle = \langle \mathbf{N}_t, \mathbf{T} \rangle - \kappa_s = 0$$

we can get that $\mathbf{N}_t = \kappa_s \cdot \mathbf{T}$.

Finally, by using the above equations and the normal angle $\theta_t = \kappa_s$ we can get that

$$\kappa_t = \theta_{st} = \theta_{ts} + \kappa^2 \cdot \theta_s = \kappa_{ss} + \kappa^3$$

■

Lemma 4.3. *For a smooth embedded solution $X : M^1 \times [0, T) \rightarrow \mathbb{R}^2$ of the curve shortening*

flow, the length $L(t)$ evolves according to the following equation:

$$\frac{dL}{dt}(t) = - \int_{M^1} \kappa^2 du \quad (4.8)$$

Proof. Let s be the arc length, then it's easy to know that for any fixed t there are

$$ds = |X_u(u, t)| du \quad (4.9)$$

Thus we have

$$\frac{dL}{dt} = \frac{d}{dt} \int_{M^1} ds = \int_{M^1} \frac{\partial}{\partial t} |X_u(u, t)| du$$

Now we are going to calculate $\partial |X_u| / \partial t$. Since $X(u, t)$ is a smooth embedded solution of the curve shortening flow, we have $(X_u)_t = (X_t)_u$, then there are

$$\frac{\partial}{\partial t} |X_u(u, t)| = \frac{1}{|X_u|} \langle X_u, (X_u)_t \rangle = \langle \mathbf{T}, (X_t)_u \rangle$$

From (4.5) we can get that

$$\langle \mathbf{T}, (X_t)_u \rangle = -\kappa_u \cdot \langle \mathbf{T}, \mathbf{N} \rangle - \kappa \cdot \langle \mathbf{T}, \mathbf{N}_u \rangle$$

Finally, by using $\langle \mathbf{T}, \mathbf{N} \rangle = 0$ and the Frenet-Serret equations we can get that

$$\frac{\partial}{\partial t} |X_u(u, t)| = -\kappa^2 \quad (4.10)$$

and that completes the proof. ■

Lemma 4.4. For an embedded solution $X : M^1 \times [0, T) \rightarrow \mathbb{R}^2$ to the curve shortening flow, the area $A(t)$ enclosed by $\mathcal{M}$ evolves according to the following equation:

$$\frac{dA}{dt}(t) = - \int_{M^1} \kappa ds = -2\pi \quad (4.11)$$

Proof. By using (4.9) and (4.10) we can get that

$$\frac{dA}{dt} = \frac{1}{2} \int_{M^1} (\langle X_t, \mathbf{N} \rangle + \langle X, \mathbf{N}_t \rangle) ds - \frac{\kappa^2}{2} \int_{M^1} \langle X, \mathbf{N} \rangle ds$$

From Lem.4.2 we can get that $\mathbf{N}_t = \kappa_s \cdot \mathbf{T}$, thus there are

$$\begin{aligned} \int_{M^1} (\langle X_t, \mathbf{N} \rangle + \langle X, \mathbf{N}_t \rangle) \, ds &= \int_{M^1} (\kappa_s \langle X, \mathbf{T} \rangle - \kappa) \, ds \\ &= -\kappa \int_{M^1} (\langle X_s, \mathbf{T} \rangle + \langle X, \mathbf{T}_s \rangle) \, ds - \int_{M^1} \kappa \, ds \\ &= -2 \int_{M^1} \kappa \, ds + \kappa^2 \int_{M^1} \langle X, \mathbf{N} \rangle \, ds \end{aligned}$$

and that gives

$$\frac{dA}{dt} = - \int_{M^1} \kappa \, ds = -2\pi$$

■

Now by Lem.4.3 and Lem.4.4 we can deduce the evolution of Q_α for an embedded closed curve α along the curve shortening flow, which is given by the following equation:

$$\frac{dQ_\alpha}{dt} = 8\pi^2 - 2L \int_{M^1} \kappa^2 \, ds$$

Then by using the Cauchy-Schwarz inequality we can get that

$$\frac{dQ_\alpha}{dt} = 8\pi^2 - 2 \left(\int_{M^1} 1^2 \, ds \right) \left(\int_{M^1} \kappa^2 \, ds \right) \leq 8\pi^2 - 2 \left(\int_{M^1} \kappa \, ds \right)^2 = 0$$

This shows that Q_α is indeed monotonically non-increasing along the curve shortening flow.

4.1.2 Curvature Estimation

Now we are going to compute the limit of Q_α as $t \rightarrow T$. To do that, we first need to estimate the curvature κ along the curve shortening flow which will show that both the length $L(t)$ and area $A(t)$ will converge to zero as $t \rightarrow T$.

Proposition 4.1. *If $X_0 : \mathbb{R}/AZ \rightarrow \mathbb{R}^2$ is an immersion, then the maximal time of existence of the curve shortening flow with initial data X_0 is finite which we denote by T .*

Proof. Let $X : \mathbb{R}/AZ \rightarrow \mathbb{R}^2$ be a solution to the curve shortening flow with initial data X_0 . Since $\mathbb{R}/AZ$ is compact, the image of X_0 is contained in some large ball $B_R(0)$ in $\mathbb{R}^2$. In particular X_0 is disjoint from the circle $\tilde{X}_0(\theta)$. By the avoidance principle [1] this remains true for positive times, and so $X(\mathbb{R}/AZ, t)$ is enclosed by the circle given by the image of the map $\tilde{X}(\theta, t) = \sqrt{R^2 - 2t} \cdot (\cos \theta, \sin \theta)$. It's easy to verify that $\tilde{X}(\theta, t)$ is a solution to the curve shortening flow, and thus the maximal time of existence of X is at most $R^2/2$. ■

Our approach to determining the bounds of the curvature κ proceeds as follows: First, we observe that a point x on a smooth curve where the curvature κ is large, can be viewed locally as a part of

circle of radius $1/\kappa$. Consequently—drawing upon classical results from Euclidean geometry—we readily deduce that if s denotes the arc length, the distance d between any two points would have the following approximation:

$$d(x, y) \approx \frac{2}{\kappa} \cdot \sin\left(\frac{\kappa \cdot s}{2}\right) \approx z - \frac{\kappa^2}{24} \cdot z^3$$

Thus, provided we can identify a function φ such that $\varphi(x) \geq x - Cx^3 + o(x^3)$ as $x \rightarrow 0$ and $d(x, y) \geq \varphi(s(x, y))$, then we can derive an upper-bound estimate for the curvature of the form $\kappa^2 \leq 24C$. We now proceed to elaborate on this line of reasoning in detail.

Definition 4.1. Let $X : M^1 \rightarrow \mathbb{R}^2$ be a smooth embedding. The cord-arc profile of X is a function $\psi_X : [0, \infty) \rightarrow \mathbb{R}$ defined by

$$\psi_X(z) = \inf \{|X(x) - X(y)| : s(x, y) = z\} \quad (4.12)$$

where $s(x, y)$ is the arc length between x and y along X .

We also define the extrinsic distance $d : M^1 \times M^1 \times [0, T) \rightarrow \mathbb{R}$ on such embedded solution X as

$$d(x, y, t) = |X(x, t) - X(y, t)|$$

Then by using the Huisken's distance comparison estimate, we can get the following proposition.

Proposition 4.2. Let $X : M^1 \rightarrow \mathbb{R}^2$ be a smooth embedding and let $K = \sup_{M^1} |\kappa|$. Then

$$\psi_X(z) \geq z - \frac{K^2}{24} \cdot z^3 + O(z^5), \quad z \rightarrow 0 \quad (4.13)$$

Proof. We first claim that for any (x, y) with $s(x, y) \leq \pi/K$, there are

$$d(x, y) \geq \frac{2}{K} \cdot \sin\left[\frac{K \cdot s(x, y)}{2}\right]$$

Then by using the Taylor expansion of $\sin z$ at $z = 0$ we can get that

$$\psi_X(z) \geq z - \frac{K^2}{24} \cdot z^3 + O(z^5), \quad z \rightarrow 0$$

To prove this claim, we let x and y be two points on the curve joined by a curve segment of length $s(x, y) \leq \pi/K$. Let u be an arc length parameter with $u(x) = -s/2$ and $u(y) = s/2$. For any $u \in [-s/2, s/2]$ we have

$$|\theta(u) - \theta(0)| \leq \left| \int_0^u \kappa \, du \right| \leq K |u| \leq \frac{K \cdot s(x, y)}{2} \leq \frac{\pi}{2}$$

Then note that

$$\begin{aligned}\langle X(y) - X(x), \mathbf{T}(0) \rangle &= \int_{-s/2}^{s/2} \langle \mathbf{T}(u), \mathbf{T}(0) \rangle \, du \\ &\geq \int_{-s/2}^{s/2} \cos(K|u|) \, du = \frac{2}{K} \cdot \sin \left[\frac{K \cdot s(x, y)}{2} \right]\end{aligned}$$

By using the following Cauchy-Schwarz inequality one can see that the claim holds.

$$|X(y) - X(x)| \geq \langle X(y) - X(x), \mathbf{T}(0) \rangle$$

To get the reverse inequality we will show that for any small z there exists points (x, y) with $s(x, y) = z$ and

$$d(x, y) \leq z - \frac{K^2}{24} \cdot z^3 + C \cdot z^5$$

for some constant C . Now we first consider the Tylor expansion of such embedding X with the arc length parameter u at the point $u = 0$:

$$X(u) = X(0) + \sum_{n=1}^4 \frac{u^n}{n!} \cdot \frac{d^n X}{du^n}(0) + O(u^5), \quad u \rightarrow 0$$

By the Frenet-Serret equations we know that $X'(0) = \mathbf{T}$ and $X''(0) = -\kappa \mathbf{N}$. Also by computing their derivatives we can get that

$$X'''(0) = -\kappa' \mathbf{N} - \kappa^2 \mathbf{T}, \quad X^{(4)}(0) = (\kappa^3 - \kappa'') \mathbf{N} - 3\kappa \kappa' \mathbf{T}$$

This gives that

$$X(u) - X(0) = u \mathbf{T} - \frac{u^2}{2} \kappa \mathbf{N} - \frac{u^3}{6} (\kappa' \mathbf{N} + \kappa^2 \mathbf{T}) + \frac{u^4}{24} [(\kappa^3 - \kappa'') \mathbf{N} - 3\kappa \kappa' \mathbf{T}] + O(u^5)$$

Then for $s \rightarrow 0$, we can deduce that

$$\left| X\left(\frac{s}{2}\right) - X\left(-\frac{s}{2}\right) \right|^2 = s^2 - \frac{s^4}{12} \kappa^2 + O(s^6)$$

Finally, by considering the Taylor expansion of $\sqrt{1+x}$ at $x = 0$ we can get that

$$d\left(\frac{s}{2}, -\frac{s}{2}\right) = \left| X\left(\frac{s}{2}\right) - X\left(-\frac{s}{2}\right) \right| = s - \frac{s^3}{24} \kappa^2 + O(s^5), \quad s \rightarrow 0$$

This completes the proof. ■

Now we are going to find the function φ such that $\varphi(x) \geq x - Cx^3 + o(x^3)$ as $x \rightarrow 0$ and $d(x, y) \geq \varphi(s(x, y))$. A natural way to do that is to study the condition on φ which will guarantee the positivity of the function $d(x, y) - \varphi(s(x, y))$. Thus we define the function $Z : M^1 \times M^1 \times [0, T] \rightarrow \mathbb{R}$

as

$$Z(x, y, t) = d(x, y, t) - L(t) \cdot \varphi \left[\frac{s(x, y, t)}{L(t)}, t \right]$$

where $\varphi(x, t) = \varphi_t(x)$ for each $t \in [0, T]$. Obviously, Z is well-defined and smooth. So there are

$$\frac{\partial Z}{\partial x} = \frac{\partial d}{\partial x} - \varphi'_t \cdot \frac{\partial s}{\partial x}, \quad \frac{\partial Z}{\partial y} = \frac{\partial d}{\partial y} - \varphi'_t \cdot \frac{\partial s}{\partial y}$$

By letting $w = R/d = [X(x) - X(y)]/|X(x) - X(y)|$ and considering the differentiation of $d^2 = \langle X(x) - X(y), X(x) - X(y) \rangle$ we get that

$$\frac{\partial d}{\partial x} = -\langle w, \mathbf{T}(x) \rangle, \quad \frac{\partial d}{\partial y} = \langle w, \mathbf{T}(y) \rangle$$

Since $\partial s / \partial x = -1$ and $\partial s / \partial y = 1$, we have

$$\frac{\partial Z}{\partial x} = -\langle w, \mathbf{T}(x) \rangle + \varphi'_t, \quad \frac{\partial Z}{\partial y} = \langle w, \mathbf{T}(y) \rangle - \varphi'_t$$

By computing the second derivatives of d we can get the second derivatives of Z as follows:

$$\begin{aligned} \frac{\partial^2 Z}{\partial x^2} &= \langle w, \kappa(x) \mathbf{N}(x) \rangle + \frac{1}{d} \langle \mathbf{T}(x) - \langle w, \mathbf{T}(x) \rangle w, \mathbf{T}(x) \rangle - \frac{\varphi''_t}{L(t)} \\ \frac{\partial^2 Z}{\partial y^2} &= -\langle w, \kappa(y) \mathbf{N}(y) \rangle + \frac{1}{d} \langle \mathbf{T}(y) - \langle w, \mathbf{T}(y) \rangle w, \mathbf{T}(y) \rangle - \frac{\varphi''_t}{L(t)} \\ \frac{\partial^2 Z}{\partial x \partial y} &= -\frac{1}{d} \langle \mathbf{T}(x) - \langle w, \mathbf{T}(x) \rangle w, \mathbf{T}(y) \rangle + \frac{\varphi''_t}{L(t)} \end{aligned}$$

Combined with the equation of curve shortening flow (4.5) and the result of Lem.4.3 we can also get the time derivative of Z as follows:

$$\begin{aligned} \frac{\partial Z}{\partial t} &= \frac{\partial d}{\partial t} - \frac{dL}{dt} \cdot \varphi - \frac{\varphi'_t}{L(t)} \left(\frac{\partial s}{\partial t} \cdot L - \frac{dL}{dt} \cdot s \right) - \frac{\partial \varphi}{\partial t} \cdot L(t) \\ &= \langle w, \kappa(x) \mathbf{N}(x) - \kappa(y) \mathbf{N}(y) \rangle + \left(\varphi - \frac{s}{L} \varphi'_t \right) \int_{M^1} \kappa^2 ds + \varphi'_t \int_x^y \kappa^2 ds - \frac{\partial \varphi}{\partial t} \cdot L(t) \end{aligned}$$

Next, we consider all possible local minima of Z . A necessary condition for Z to be strictly positive is evidently that its values at these points are strictly positive. Based on the necessary conditions for local extrema, we set all first-order partial derivatives of Z to zero to obtain

$$\langle w, \mathbf{T}(x) \rangle = \langle w, \mathbf{T}(y) \rangle = \varphi'_t$$

This leads to two possibilities: Either the two tangent directions $\mathbf{T}(x)$ and $\mathbf{T}(y)$ are parallel, or they are bisected by the direction w . We can eliminate one of these possibilities by assuming that φ is concave, that is Prop.4.3. Then by hypothesis $Z \geq 0$ we know that for a concave φ the only possibility is the case where $\mathbf{T}(x)$ and $\mathbf{T}(y)$ are bisected by w .

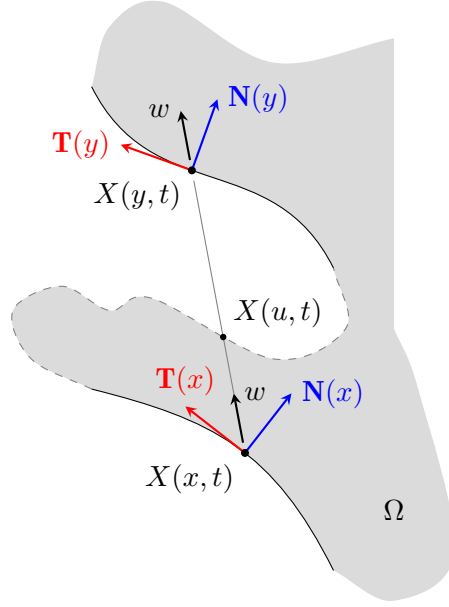


Figure 4.1: For the case of $\mathbf{T}(x) = \mathbf{T}(y)$ there must be a point $X(u, t)$ on the curve along the line joining the line segment from $X(x, t)$ to $X(y, t)$ and for any concave φ it follows that (x, y) does not minimize Z .

Proposition 4.3. *Suppose that $\varphi(x)$ is symmetric about $z = 1/2$ and is strictly concave with $|\varphi'_t| < 1$ on $(0, 1)$, and let $(x, y) \in (M^1 \times M^1) \setminus \{(x, x) \mid x \in M^1\}$ be a critical point of Z with $\mathbf{T}(x) = \mathbf{T}(y)$ and $Z(x, y, t) = 0$. Then there exists $(u, v) \in M^1 \times M^1$ such that $Z(u, v, t) < 0$.*

Proof. According the Jordan curve theorem, the closed curve X separates $\mathbb{R}^2$ into precisely two open connected regions. Then as $\langle w, \mathbf{T}_x \rangle = \varphi' \in (-1, 1)$ we know that w and $\mathbf{T}_x$ are linearly independent. It follows that for small $a > 0$, the point $X(x, t) + aw$ lies in one of two connected components of $\mathbb{R}^2 \setminus X(M^1)$, we denote it as Ω . By carefully choosing the direction of parametrization, Ω can also contain the point $X(x, t) + a\mathbf{N}(x)$ for some small $a > 0$ (as shown in Fig.4.1), since φ is symmetric about $z = 1/2$. Thus with the connectedness, Ω should also contain points $X(z, t) + a\mathbf{N}(z)$ for any $z \in M^1$ and so as point $X(y, t) + a\mathbf{N}(y)$. Then by $\mathbf{T}(x) = \mathbf{T}(y)$ we know that $\mathbf{N}(x) = \mathbf{N}(y)$ which implies that Ω contains point $X(y, t) + aw$ for some small $a > 0$ but never contain any point $X(y, t) - aw$ with $a \in (0, \epsilon)$ (as shown in Fig.4.1).

Let $d = d(x, y, t)$ then there exists $d^* > 0$ such that

$$d^* = \sup \{s \in (0, d) \mid X(x, t) + sw \in \Omega\}$$

and from the above discussion we know that $0 < d^* < d$ and $X(x, t) + d^*w \in \partial(\Omega) = X(M^1)$. Hence there exists $u \in M^1$ such that

$$X(u, t) = X(x, t) + d^*w$$

Then as shown in Fig.4.1 we have

$$d(x, y, t) = d(x, u, t) + d(u, y, t), \quad s(x, y, t) = s(x, u, t) + s(u, y, t)$$

Therefore, by the strict concavity we can get that

$$\varphi \left[\frac{s(x, u, t)}{L(t)}, t \right] + \varphi \left[\frac{s(u, y, t)}{L(t)}, t \right] > \varphi \left[\frac{s(x, y, t)}{L(t)}, t \right]$$

Since $Z(x, y, t) = 0$, we know that

$$\varphi \left[\frac{s(x, y, t)}{L(t)}, t \right] = \frac{d(x, y, t)}{L(t)} = \frac{d(x, u, t)}{L(t)} + \frac{d(u, y, t)}{L(t)}$$

Thus we get that

$$\varphi \left[\frac{s(x, u, t)}{L(t)}, t \right] + \varphi \left[\frac{s(u, y, t)}{L(t)}, t \right] > \frac{d(x, u, t)}{L(t)} + \frac{d(u, y, t)}{L(t)}$$

It follows that either $\varphi[s(x, u, t)/L, t] > d(x, u, t)/L$ or $\varphi[s(u, y, t)/L, t] > d(u, y, t)/L$, which means at least one of $Z(x, u, t)$ and $Z(u, y, t)$ is negative. So we complete the proof. ■

Since $\mathbf{T}(x)$ and $\mathbf{T}(y)$ are bisected by w , we let θ be the angle between $\mathbf{T}(x)$ and w , then it equals the angle between $\mathbf{T}(y)$ and w . Now we are going to prove the following proposition (Prop.4.4) which states that if a strictly concave smooth function φ with $|\varphi'| \leq 1$ also satisfies the inequality (4.14) then the minimal value of Z can occur only at the instant $t = 0$; consequently, ensuring that Z is non-negative at every point at the initial moment suffices to guarantee its positivity. Furthermore, the resulting Inequality (4.14) along with other supplementary conditions can, in turn, assist us in determining the structure of the function φ .

Proposition 4.4. *Let $X : \mathcal{M}^1 \times [0, T) \rightarrow \mathbb{R}^2$ be an embedded solution of curve shortening flow (4.5) and let $\varphi : [0, 1] \times [0, \infty) \rightarrow \mathbb{R}$ be a smooth nonnegative function which is symmetric about $z = 1/2$, strictly concave in its first argument, and satisfies $|\varphi'| \leq 1$. Suppose that φ satisfies the inequality*

$$\frac{\partial \varphi}{\partial \tau} \leq 4(\varphi'' + \pi^2 \varphi) + \frac{8\pi \varphi'}{\tan(\pi z)} - \frac{8\pi(\varphi')^2}{\sin(\pi z)} \quad (4.14)$$

at any point with $|\varphi'| \leq 1$. If

$$d(x, y, 0) \geq L(0) \cdot \varphi \left[\frac{s(x, y, 0)}{L(0)}, 0 \right] \quad (4.15)$$

for all $x, y \in \mathcal{M}^1$, then the following inequality holds for any $(x, y, t) \in \mathcal{M}^1 \times [0, T)$, where

$$\tau(t) = \int_0^t L^{-2}(t) dt.$$

$$d(x, y, t) \geq L(t) \cdot \varphi \left[\frac{s(x, y, t)}{L(t)}, \tau(t) \right]$$

Specifically, we will prove this proposition by using the method of proof by contradiction. Suppose that there exists $(x, y, t_0) \in \mathcal{M}^1 \times (0, T)$ is the first point that $Z = 0$ then for any $(x, y, t) \in \mathcal{M}^1 \times (0, t_0)$ there are $Z(x, y, t) > 0$. By the assumption $Z \geq 0$ we know that (x, y, t_0) is a local minimum point of Z then we should have

$$\left[\frac{\partial}{\partial t} - \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y} \right)^2 \right] (Z) \Big|_{(x, y, t_0)} \leq 0$$

However, we will show that for the function φ that satisfies (4.14) and other such conditions there are

$$\left[\frac{\partial}{\partial t} - \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y} \right)^2 \right] (Z) > 0 \quad (4.16)$$

for any $(x, y, t) \in \mathcal{M}^1 \times [0, T)$, which is a contradiction. Thus, the point $Z(x, y, t_0) = 0$ can only exist when $t_0 = 0$. By $Z(\cdot, 0) \geq 0$ we finally could get $Z(x, y, t) \geq 0$ for any $(x, y, t) \in \mathcal{M}^1 \times [0, T)$ and that completes the proof.

Lemma 4.5. *If φ_t is strictly concave and positive on $[0, 1]$ and $\varphi(z, t) = \varphi(1 - z)$, then for all $z \in [0, 1/2)$ there are $\varphi'_t > 0$ and*

$$\varphi_t - z\varphi'_t > 0 \quad (4.17)$$

Proof. Since $\varphi(z, t) = \varphi(1 - z)$ for any $z \in [0, 1]$ then we have

$$\varphi'(z, t) = -\varphi'(1 - z, t)$$

For $z = 1/2$ we have that $2\varphi'(1/2, t) = 0$ which means $z = 1/2$ is a critical point. Hence, by the strict concavity, there are $\varphi > 0$ and $\varphi' > 0$ for any $z < 1/2$. Now consider the function $u(z) = \varphi - z\varphi'$. It's easy to know that $u(0) = \varphi(0, t) > 0$ and

$$u'(z) = \varphi' - \varphi' - z\varphi''(z) > 0$$

It follows that $u > 0$ on $(0, 1]$ ■

Note that we also have the following two Cauchy-Schwarz inequalities for the related integrals.

$$\begin{aligned} s(x, y, t) \int_x^y \kappa^2 ds &= \left(\int_x^y 1^2 ds \right) \cdot \int_x^y \kappa^2 ds \geq \left(\int_x^y \kappa ds \right)^2 = 4\theta^2 \\ L(t) \int_{M^1} \kappa^2 ds &= \left(\int_{M^1} 1^2 ds \right) \cdot \int_{M^1} \kappa^2 ds \geq \left(\int_{M^1} \kappa ds \right)^2 = 4\pi^2 \end{aligned}$$

Then by Lem.4.5 and the result of the previous calculations of the derivatives of Z we have the following estimate for $[\partial_t - (\partial_x - \partial_y)^2](Z)$.

$$\begin{aligned} \left[\frac{\partial}{\partial t} - \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y} \right)^2 \right] (Z) &= -L(t) \frac{\partial \varphi}{\partial t} + 4 \frac{\varphi''}{L(t)} + \varphi' \int_x^y \kappa^2 ds + \left[\varphi - \frac{s(x, y, t)}{L(t)} \cdot \varphi' \right] \int_{M^1} \kappa ds \\ &> -L(t) \frac{\partial \varphi}{\partial t} + 4 \frac{\varphi''}{L(t)} + \frac{4\theta^2}{s(x, y, t)} + \frac{4\pi^2}{L(t)} \left(\varphi - \frac{s}{L} \varphi' \right) \\ &= -L(t) \frac{\partial \varphi}{\partial t} + 4 \frac{\varphi''}{L(t)} + \frac{4 \arccos^2(\varphi')}{s(x, y, t)} + \frac{4\pi^2}{L(t)} \left(\varphi - \frac{s}{L} \varphi' \right) \end{aligned}$$

where the terms involving inverse trigonometric functions arise from the vanishing of the first derivatives.

$$\cos \theta = \langle w, T(x) \rangle = \langle w, T(y) \rangle = \varphi'$$

The strict inequality in the derivation above stems from the fact that the condition for equality in the Cauchy inequality requires the curvature κ of the curve to be constant everywhere; however, the assumption that the curve is not a circle precludes this scenario. (This condition for equality is, in fact, precisely the source of the equality condition for the isoperimetric inequality that we ultimately derived.)

Let $z = s/L \in [0, 1]$ then (4.16) holds if φ satisfies the following inequality

$$L^2(t) \cdot \frac{\partial \varphi}{\partial t} \leq 4\varphi'' + \frac{4}{z} \cdot \varphi' \arccos^2(\varphi') + 4\pi^2(\varphi - z\varphi') \quad (4.18)$$

From this to get (4.14) we need to further process the terms in (4.18) that contain L^2 as well as inverse trigonometric functions.

Lemma 4.6. *The function $f(x) = \arccos^2(x)$ is convex on $[0, 1]$, and we also have the following estimation for any $x, z \in [0, 1]$.*

$$f(x) \geq \pi^2 z^2 + \frac{2\pi z}{\sin(\pi z)} [\cos(\pi z) - x] \quad (4.19)$$

Proof. Let $y = f(x)$ then one can easily get that $y' = -2\sqrt{y}/\sin(\sqrt{y})$ and thus

$$y'' = \frac{2 \cos(\sqrt{y})}{\sin^3(\sqrt{y})} [\tan(\sqrt{y}) - \sqrt{y}] > 0$$

This shows that f is indeed convex on $[0, 1]$. It follows that

$$f(x) - f[\cos(\pi z)] \geq f'[\cos(\pi z)] \cdot [x - \cos(\pi z)]$$

Then by substituting $\cos(\pi z)$ into $f(x)$ and $f'(x)$ we get (4.19). ■

By using this Lemma and considering the following chain rule of $\tau(t) = \int_0^t L^{-2}(t) dt$

$$\frac{\partial \varphi}{\partial t} = \frac{\partial \varphi}{\partial \tau} \cdot \frac{d\tau}{dt} = \frac{1}{L^2(t)} \cdot \frac{\partial \varphi}{\partial \tau}$$

from (4.14) we can deduce that

$$\begin{aligned} L^2(t) \cdot \frac{\partial \varphi}{\partial t} &\leq 4\varphi'' + 4\pi^2\varphi + 8\pi \cdot \frac{\varphi'}{\sin(\pi z)} [\cos(\pi z) - \varphi'] \\ &= 4\varphi'' + 4\pi^2 z \varphi' + 8\pi \cdot \frac{\varphi'}{\sin(\pi z)} [\cos(\pi z) - \varphi'] + 4\pi^2 (\varphi - z\varphi') \\ &= 4\varphi'' + \frac{4\varphi'}{z} \cdot \left[\pi^2 z^2 + \frac{2\pi z}{\sin(\pi z)} [\cos(\pi z) - x] \right] + 4\pi^2 (\varphi - z\varphi') \\ &\leq 4\varphi'' + \frac{4}{z} \cdot \varphi' \arccos^2(\varphi') + 4\pi^2 (\varphi - z\varphi') \end{aligned}$$

This is exactly (4.18) and therefore Prop.4.4 has been proved.

Now we are going to find a relatively simple smooth function φ that satisfies conditions given by Prop.4.4. We construct such φ as $\varphi(z, \tau) = A(\tau) \cdot \Psi[B(\tau)C(z)]$ where A , B , C and Ψ are all smooth functions. Suppose that $C(0) = 0$ then we have

$$\varphi'(0, \tau) = A(\tau)B(\tau) \cdot \Psi'(0)C'(0)$$

We desire $\varphi'(0, \tau) = 1$, and therefore stipulate that

$$A(\tau)B(\tau) = \Psi'(0) = C'(0) = 1$$

Since it's easy to know that

$$\begin{aligned} \frac{\partial \varphi}{\partial \tau} &= A'(\tau)\Psi'(BC) + A(\tau)\Psi'(BC) \cdot B'C \\ \varphi'(z) &= AB \cdot \Psi'(BC) \cdot C' \\ \varphi''(z) &= AB \cdot C''\Psi' + AB^2(C')^2\Psi'' \end{aligned}$$

then by substituting $\varphi = A\Psi(BC)$ into (4.14) we can get that

$$\begin{aligned} A'(\tau)\Psi'(BC) + A(\tau)\Psi'(BC) \cdot B'C &\leq 4 \left[AB \cdot C''\Psi' + AB^2(C')^2\Psi'' + \pi^2 A \cdot \Psi(BC) \right] \\ &\quad + \frac{8\pi ABC'\Psi'}{\tan(\pi z)} - \frac{8\pi(ABC'\Psi')^2}{\sin \pi z} \end{aligned}$$

From $AB = 1$ we know that $B = A^{-1}$ and it follows that $AB' = -A'/A$, $AB^2 = 1/A$ and $(AB)^2 = 1$. Substituting those results into the above inequality we can get that

$$\begin{aligned} A'\Psi - \frac{CA'}{A}\Psi' &\leq \frac{4(C')^2}{A}\Psi'' + 4C''\Psi' + 4\pi^2 A\Psi + \frac{8\pi C'\Psi'}{\tan \pi z} - \frac{8\pi(C')^2(\Psi')^2}{\sin \pi z} \\ &= \frac{4(C')^2}{A}\Psi'' + 4C''\Psi' + 4\pi^2 A\Psi + \frac{8\pi\Psi'C'}{\tan(\pi z)} \left[1 - \frac{C'\Psi'}{\cos(\pi z)} \right] \end{aligned}$$

Now let $A = e^{4\pi^2\tau}$ and $C = \pi^{-1} \sin(\pi z)$ then we have

$$0 \leq \frac{4 \cos^2(\pi z)}{e^{4\pi^2\tau}} \left[\Psi'' + \frac{2\pi \cdot e^{4\pi^2\tau}}{\sin(\pi z)} \Psi'(1 - \Psi') \right]$$

From this we just focus on the following ODE

$$\begin{cases} \Psi'' + (2/\xi)\Psi'(1 - \Psi') = 0 \\ \Psi(0) = 0 \\ \Psi'(0) = 1 \end{cases} \quad (4.20)$$

where $\xi = \pi^{-1}e^{-4\pi^2\tau} \sin(\pi z)$ is the argument of Ψ . Let $p(z) = 1/\Psi'(z)$ then (4.20) gives that

$$p' = \frac{2}{\xi}(p - 1)$$

From this we can solve p as $p = 1 + C \cdot \xi^2$ and that means $\Psi'(\xi) = (1 + C \cdot \xi^2)^{-1}$. Therefore, we have

$$\Psi(\xi) = \Psi(0) + \int_0^\xi \frac{dz}{1 + Cz^2}$$

If $C = 0$ then we have $\Psi(\xi) = \xi$, it gives that $\varphi(z) = \pi^{-1} \sin \pi z$ which is the circle chord-arc profile.

If $C < 0$ then we have

$$\Psi(\xi) = \frac{1}{a} \operatorname{arctanh}(a\xi)$$

This time one can easily varify that such φ contradicts the condition $|\varphi'| \leq 1$. If $C > 0$ then there exists some $a \in \mathbb{R}$ such that $a^2 = C$ and that gives

$$\Psi(\xi) = \frac{1}{a} \operatorname{arctan}(a\xi)$$

Hance, in this case there are

$$\varphi(z) = \frac{e^{4\pi^2\tau(t)}}{a} \cdot \arctan \left[\frac{a}{\pi e^{4\pi^2\tau(t)}} \cdot \sin(\pi z) \right] \quad (4.21)$$

where $z = s(x, y, t)/L(t) \in [0, 1]$.

Now along with the idea we introduced at the begining of this section we are going to prove the following theorem which gives the global bound of the curvature κ , by varifying that such φ given by (4.21) satisfies conditions listed in Prop.4.4.

Theorem 4.2. *If $X : M^1 \times [0, T) \rightarrow \mathbb{R}^2$ is a smooth solution of curve shortening flow with a smooth initial embedding $X(u, 0) = X_0 : M^1 \rightarrow \mathbb{R}^2$ from a compact 1-manifold to $\mathbb{R}^2$ then for any $(x, t) \in M^1 \times [0, T)$ the curvature κ have that*

$$\kappa^2(x, t) \leq \frac{4\pi^2}{L^2(t)} \cdot \left[1 + \frac{2a^2}{\pi^2} e^{-8\pi^2\tau(t)} \right] \quad (4.22)$$

where $\tau(t) = \int_0^t L^{-2}(t) dt$.

Proof. We are first going to varify that such φ given by (4.21) satisfies all conditions in Prop.4.4. Since Ψ is the solution of the ODE (4.20) then φ naturally satisfies (4.14). By the monotonicity of $\arctan(x)$ on $[0, 1]$ and the following equility we could deduce that $\varphi(1/2 - h) = \varphi(1/2 + h)$ for any $h \in (0, 1)$.

$$\sin \pi \left(\frac{1}{2} + h \right) = \cos \pi h = \sin \pi \left(\frac{1}{2} - h \right)$$

That gives the symmetry of φ . Obviously, such φ is smooth and nonnegative. Now consider $|\varphi'|$, by the previous calculation result we have

$$|\varphi'(z)| = |AB \cdot \Psi'(BC) \cdot C'| = \left| \frac{\cos(\pi z) \pi^2 K}{\pi^2 K + a^2 \sin^2(\pi z)} \right| \leq \frac{\pi^2 K}{\pi^2 K + a^2 \sin^2(\pi z)} \leq 1$$

where $K = \pi^2 e^{8\pi^2\tau(t)}$. Finally, for the second derivative of φ we have that

$$\varphi''(z) = \frac{[(-\pi^3 K \sin(\pi z))] [\pi^2 K + a^2 \sin^2(\pi z)] - [\cos(\pi z) \cdot \pi^2 K] [2\pi a^2 \sin(\pi z) \cos(\pi z)]}{[\pi^2 K + a^2 \sin^2(\pi z)]^2}$$

Note that the numberator could be written as

$$-\pi K \sin(\pi z) [\pi^2 K (\pi^2 K + a^2 \sin^2 \pi z) + 2\pi a^2 \cos^2(\pi z)]$$

Its negativity gives that $\varphi'' < 0$ on $[0, 1]$. Thus φ is indeed strictly concave.

Now we will show that for any smooth embedding $X_0 : M^1 \rightarrow \mathbb{R}^2$ there exists a constant $a > 0$

such that

$$d(x, y, 0) \geq \frac{L(0)}{a} \cdot \arctan \left[\frac{a}{\pi} \cdot \sin \left(\frac{\pi s}{L} \right) \right] \quad (4.23)$$

and this is exactly (4.15). Therefore, by Prop.4.4 we get that for any $(x, y, t) \in M^1 \times M^1 \times [0, T]$ there are

$$d(x, y, t) \geq \frac{L(t)e^{4\pi^2\tau(t)}}{a} \cdot \arctan \left[\frac{a}{\pi e^{4\pi^2\tau(t)}} \cdot \sin \left(\frac{\pi s}{L} \right) \right] \quad (4.24)$$

Let $L = L(0)$ and define a function $\Phi_a : [0, L/2] \rightarrow \mathbb{R}$ for some $a > 0$ as follows

$$\Phi_a(s) = \frac{L}{a} \cdot \arctan \left[\frac{a}{\pi} \cdot \sin \left(\frac{\pi s}{L} \right) \right]$$

It's easy to know that for any $(x, x) \in M^1$ and $a > 0$ there are

$$\Phi_a[s(x, x, 0)] = \Phi_a(0) = 0 = d(x, x, 0)$$

Thus we should only focus on such points $(x, y) \in M^1$ with $x \neq y$. Now we fix s as some values in $[0, L/2]$ and let $r = \pi^{-1} \sin(\pi s/L)$ then there are

$$\Phi_a(s) = \frac{L}{a} \cdot \arctan(ar)$$

Easily one can find the derivative of $\Phi_a(s)$ on a as

$$\frac{\partial \Phi_a(s)}{\partial a} = \frac{1}{a^2} \left[\frac{ar}{1 + a^2 r^2} - \arctan(ar) \right]$$

Since it's also well known that for any $z > 0$ there are

$$\arctan(z) = \int_0^z \frac{1}{1 + u^2} du > \frac{z^2}{1 + z^2}$$

Then by the negativity of $\partial_a(\Phi_a)$ we know that $\Phi_a(s)$ is strictly decreasing with respect to a for any $s \in [0, L/2]$. Let $z = s(x, y, 0)$ be the arc length between x and y on the initial curve X_0 and $\mathbf{T}_0(s)$ be the tangent vector of X_0 at point $X_0(s)$. Since the curve X_0 is smooth and M^1 is compact then it's well known that there exists $K > 0$ such that

$$K = \max_{M_1} |\kappa_0| < \infty$$

where $\kappa_0 = \kappa(s, 0)$ is the curvature of X_0 with the arc length parameter s . By the Taylor expansion we know that as $z \rightarrow 0$ for any fixed $a > 0$ there are

$$\Phi_a(z) = z - \frac{\pi^2 + 2a^2}{6L^2} \cdot z^3 + O(z^5)$$

Since $\lim_{a \rightarrow +\infty} (\pi^2 + 2a^2)/(6L^2) = +\infty$ then there exists $a_0 > 0$ such that $(\pi^2 + 2a_0^2)/(6L^2) > K^2/6$.

Hence for some sufficiently small $\delta > 0$ there are

$$\Phi_{a_0}(z) \leq z - \frac{K^2}{6}z^3, \quad \forall z \in (0, \delta) \quad (4.25)$$

Now for some $z \in (0, \delta)$ we take $x = X_0(s)$ and then let $y = X_0(s + z)$. By the Cauchy-Schwarz inequality we have

$$d(x, y, 0) = |X_0(s + z) - X_0(s)| \geq \langle X_0(s + z) - X_0(s), \mathbf{T}_0(s) \rangle = \int_0^z \langle \mathbf{T}(s + \xi), \mathbf{T}(s) \rangle d\xi$$

Since $|\mathbf{T}_0| = |\kappa| \leq K$ then by the Lagrangian mean value theorem we have that for any $\xi \in (0, z)$ there are

$$|\mathbf{T}(s + \xi) - \mathbf{T}(s)| \leq K \cdot \xi$$

Therefore we have the following estimation of $d(x, y, 0)$:

$$\begin{aligned} d(x, y, 0) &\geq \int_0^z \langle \mathbf{T}(s + \xi), \mathbf{T}(s) \rangle d\xi \\ &= \int_0^z \left(1 - \frac{1}{2} |\mathbf{T}(s + \xi) - \mathbf{T}(s)|^2 \right) d\xi \geq \int_0^z \left(1 - \frac{1}{2} K^2 \xi^2 \right) d\xi = z - \frac{K^2}{6} z^3 \end{aligned}$$

This time, by (4.25) we get that for any $z \in (0, \delta)$ there are $d(x, y, 0) \geq \Phi_{a_0}(z)$. Now let's consider such $(x, y) \in M^1 \times M^1$ with $z = s(x, y, 0) \geq \delta$. Let A_δ be the set of such (x, y) which is defined as

$$A_\delta = \{(x, y) \in M^1 \times M^1 \mid s(x, y, 0) \geq \delta, x \neq y\}$$

By the compactness of M^1 we know that A_δ is also compact. Since X_0 is a smooth embedding then for any $(x, y) \in A_\delta$ there are $X_0(x) \neq X_0(y)$ which gives that

$$d(x, y, 0) > 0, \quad \forall (x, y) \in A_\delta$$

Thus there exists $m > 0$ such that for any $(x, y) \in A_\delta$ there are $d(x, y, 0) \geq m$. Since $\pi L/2a \rightarrow 0$ as $a \rightarrow +\infty$, then there exists $a_1 > 0$ such that $0 < \pi L/(2a_1) < m$. Hence we get that

$$\Phi_{a_1}(z) = \frac{L}{a_1} \cdot \arctan[a_1 r(z)] \leq \frac{\pi L}{2a_1} < m$$

It follows that for any $z \geq \delta$ there are $d(x, y, 0) > \Phi_{a_1}(z)$. Finally, let $a = a_1$ then combined with the monotonicity of Φ_a we have that

$$\Phi_a(s(x, y, 0)) = \Phi_{a_1}(z) \leq d(x, y, 0)$$

and that is exactly (4.23).

Ultimately, with the Taylor expansion of φ given by (4.21) we have

$$\varphi(z) = z - \left(\frac{\pi^2}{6} + \frac{a^2}{3e^{8\pi^2\tau(t)}} \right) z^3 + O(z^5)$$

as $z \rightarrow 0$. Then combined with Prop.4.2 we can get that

$$K^2 \leq 24 \left(\frac{\pi^2}{6} + \frac{a^2}{3e^{8\pi^2\tau(t)}} \right) = \frac{4\pi^2}{L^2(t)} \cdot \left[1 + \frac{2a^2}{\pi^2} e^{-8\pi^2\tau(t)} \right]$$

and we complete the proof. ■

4.1.3 Global Existence and the Limit Calculation

In this section, we will give a sketch of the proof for Thm.4.3; by leveraging this result in conjunction with Thm.4.2—introduced in the previous section—we will derive Conclusion of $\lim_{t \rightarrow T} L(t) = 0$. Finally, through a detailed proof of Lem.4.9, we will establish

$$\lim_{t \rightarrow T} Q_\alpha(t) = \lim_{t \rightarrow T} [L^2(t) - 4\pi \cdot A(t)] = 0$$

thereby completing the proof of the isoperimetric inequality (Thm.4.1) for the case of smooth embedded planar curve.

Theorem 4.3. *Let $X : M^1 \times [0, T) \rightarrow \mathbb{R}^2$ be a smooth solution of the curve shortening flow. If T is the maximal time of existence then*

$$\limsup_{t \rightarrow T} \sup_{M^1 \times \{t\}} |\kappa| = \infty \quad (4.26)$$

We will prove this theorem by using the proof of contrapositive. Specifically, we will show that if there are

$$\sup_{M^1 \times [0, T)} |\kappa| = K < \infty$$

then we can find some $\delta > 0$ such that X extends smoothly to $M^1 \times [0, T + \delta)$ which is contradicted to Prop.4.1. To do this we will first find some bounds for the parameterization map X and its derivatives, then by applying the Arzela-Ascoli theorem we will obtain a smooth limit solution as $t \rightarrow T$ and thereby extend the solution via the following theorem (the detailed proof of this theorem can be found in [2])

Theorem 4.4. *Let M^1 be a closed 1-manifold and let $X_0 : M^1 \rightarrow \mathbb{R}^2$ be a smooth immersion. Then there exists $\epsilon > 0$ and a smooth solution $X(t) : M^1 \rightarrow \mathbb{R}^2$ to the curve shortening flow defined for $t \in [0, \epsilon)$ and satisfying $X(0) = X_0$.*

Firstly, one can find a very simple supremum bound:

$$|X(x, t) - X(x, 0)| \leq \int_0^t \left| \frac{\partial X}{\partial t}(x, \tau) \right| d\tau \leq K|t| \leq KT$$

Now we fix a parameter x as the arc length parameter for the curve when $t = 0$. Then we are going to control all of the derivatives of X with respect to this fixed parametrization.

Lemma 4.7. *For any $x \in M^1$ and $t \in [0, T]$, there are*

$$\exp(-K^2T) \leq |X'(x, t)| \leq 1 \quad (4.27)$$

Proof. We first claim that for any solution $X(u, t)$, X' denotes its first derivative of u that is $X' = \partial_x X$. Then by differentiate on the both sides of $|X'|^2 = \langle X', X' \rangle$ we can get that

$$\frac{\partial}{\partial t} |X'| = \frac{1}{|X'|} \cdot \left\langle X', \frac{\partial X'}{\partial t} \right\rangle = \left\langle \mathbf{T}, \frac{ds}{dx} \cdot \frac{\partial}{\partial s} \cdot \frac{\partial X}{\partial t} \right\rangle = |X'| \left\langle \mathbf{T}, \frac{\partial}{\partial s} (-\kappa \cdot \mathbf{N}) \right\rangle$$

Then combined with (4.6) we can get that

$$\frac{\partial}{\partial t} |X'| = -\kappa^2 |X'|$$

By solving this ODE we know that

$$|X'(x, t)| = \exp \left[- \int_0^t \kappa^2(x, \tau) d\tau \right]$$

then combined with $|\kappa(x, \tau)| \leq K$ we know that

$$0 \leq \int_0^t \kappa^2(x, \tau) d\tau \leq K^2 t \leq K^2 T$$

Finally, by considering the monotonicity of the exponential function we can exactly get (4.27) as follows.

$$\exp(-K^2T) \leq \exp \left(- \int_0^t \kappa^2 d\tau \right) = |X'(x, t)| \leq \exp(0) = 1$$

■

Lemma 4.8. *For any smooth solution of the curve shortening flow the length of the curve must approach zero, that is*

$$\lim_{t \rightarrow T} L(t) = 0 \quad (4.28)$$

Lemma 4.9. *Let $X : M^1 \times [0, T) \rightarrow \mathbb{R}^2$, $M^1 \cong S^1$, be a maximal embedded solution of the curve shortening flow. Then the area $A(t)$ enclosed by $\mathcal{M}_t$ is given by*

$$A(t) = 2\pi(T - t) \quad (4.29)$$

4.2 The Case on a Sphere

In this section we are going to consider the isoperimetric problem (Prob.4.1) on a sphere. This serves not only as a generalization of the planar case discussed in the previous section but also as a springboard—and a conceptual paradigm—for subsequently addressing isoperimetric problems on more general Riemann surfaces and indeed, on some higher dimensional Riemannian manifolds. Although this problem was solved long ago—and by a variety of methods (see [3])—we wish to emphasize the elegance and conciseness of the proof technique that utilizes curve-shortening flow on the sphere, as well as the unique value this approach offers in terms of geometric intuition.

Theorem 4.5. *Let S_R^2 be a sphere with radius $R > 0$, $\Omega \subseteq S_R^2$ be a bounded, closed and simply connected region on S_R^2 . We denote $A = |\Omega|$ as the area of such region and $\gamma = \partial\Omega$ as its boundary. If L is the parameter of such closed curve γ then we always have the following isoperimetric inequality.*

$$L^2 R^2 \geq A(4\pi R^2 - A) \quad (4.30)$$

Equality holds if and only if γ is a spherical cap.

We will prove this theorem by using the same method we introduced in the planar case, that is we consider a quantity $Q(t)$ which related to (4.30), then after we showing that $Q(t)$ is monotonically non-increasing for any smooth embedded curve under the curve shortening flow and $\lim_{t \rightarrow T} Q(t) = 0$ we will get Thm.4.5. One may note that this idea is actually generalizable to the application of geometric flow techniques for solving any geometric inequality.

Since the radius R acts merely as a scaling factor under similarity transformations, we may, without loss of generality, restrict our subsequent discussion to the case of the unit sphere S^2 . Furthermore, noting that the right-hand side of Inequality (4.30) is symmetric with respect to $A = 2\pi$, we may assume that any initial closed curve γ_0 always have $0 < A \leq 2\pi$; otherwise—by virtue of the Jordan Curve Theorem—we need only consider the area of the complementary region on the sphere. Thus we now set the monoton quantity of the initial closed curve γ as

$$Q_\gamma(t) = L^2(t) - A(t)[4\pi - A(t)]$$

4.2.1 The Evolution of the Monoton Quantity

For a smooth closed curve γ let $X : M^1 \rightarrow S^2$ be its smooth immersion parameterized by arc length s , where M^1 is isomorphic to the torus $\mathbb{S}^1$. Let $\mathbf{T}(s) = X'(s)$ and $\mathbf{N}(s) = -J \cdot \mathbf{T}$ be the unit tangent and normal vector fields along γ respectively, where J is the counterclockwise rotation by $\pi/2$. Then let $\kappa_g(s)$ be the geodesic curvature of γ at s , which is defined by the following equation

$$\begin{cases} \nabla_s \mathbf{T}(s) = -\kappa_g \cdot \mathbf{N} \\ \nabla_s \mathbf{N}(s) = \kappa_g \cdot \mathbf{T} \end{cases} \quad (4.31)$$

where ∇_s is Levi-Civita of the unit sphere S^2 along γ . Let $X' = \partial_u X$ and $\mathcal{M}^1 = X(M^1)$ then by $ds = |X'|_g du$ we could calculate the length of the curve as follows

$$L = \int_{\mathcal{M}^1} ds = \int_{M^1} |X'(u)|_g du$$

Now let $X : M^1 \times [0, T] \rightarrow S^2$ be a smooth map, which is an immersion for each fixed $t \in [0, T]$. We say that $X(t)$ is a solution of the curve shortening flow on the unite sphere S^2 if it satisfies that

$$\frac{\partial X}{\partial t}(u, t) = -\kappa_g \cdot \mathbf{N}(u, t) \quad (4.32)$$

As we did in the previous section we are going to find out the first variation of $L(t)$ and $A(t)$ of $X(u, t)$ under the above curve shortening flow. We first claim that

$$\frac{dL}{dt} = - \int_{M^1} \kappa_g^2 ds \quad (4.33)$$

To prove this we first note that there are

$$|X'| \cdot \frac{\partial |X'|}{\partial t} = \langle \nabla_t X', X' \rangle_g = \langle \nabla_u X_t, X' \rangle_g$$

where $X_t = \partial_t X$. This is because the Levi-Civita connection is torsion free. Then combined with (4.32) we get that

$$\begin{aligned} |X'| \cdot \frac{\partial |X'|}{\partial t} &= \langle \nabla_u (-\kappa_g \cdot \mathbf{N}), |X'| \mathbf{T} \rangle_g \\ &= -|X'|^2 \langle \nabla_s (\kappa_g \cdot \mathbf{N}), \mathbf{T} \rangle_g \\ &= -|X'|^2 \langle (\partial_s \kappa_g) \mathbf{N}, \mathbf{T} \rangle_g - |X'|^2 \langle \kappa_g (\nabla_s \mathbf{N}), \mathbf{T} \rangle_g \end{aligned}$$

Since there are $\langle \mathbf{N}, \mathbf{T} \rangle = 0$ and $\nabla_s \mathbf{N}(s) = \kappa_g \cdot \mathbf{T}$ we finally get that

$$\frac{\partial}{\partial t} |X'| = -\kappa_g^2 |X'|$$

Therefore, the first variation of the curve length $L(t)$ could be calculated as

$$\frac{dL}{dt} = \frac{d}{dt} \int_{\mathcal{M}^1} ds = \int_{M^1} \frac{\partial}{\partial t} |X'| du = - \int_{M^1} \kappa_g^2 ds$$

and that is (4.33)

5. Valuable Research Directions

5.1 Generalizing Isoperimetric Inequality

Based on the fact that mean curvature flow constitutes the gradient flow associated with surface area, it is not difficult to deduce that its lower-dimensional counterpart—curve shortening flow—serves as the gradient flow associated with curve length. Drawing an analogy to how mean curvature flow can be utilized to address the minimal surface problem, we are naturally led to wonder: could methods derived from curve shortening flow potentially offer a novel proof of the isoperimetric inequality? The answer is affirmative; in fact, the mathematician Grayson reached a similar conclusion as early as 1983 [4]. Our task, therefore, is to clearly distill this proof from his paper and the findings derived from other relevant literature. Specifically, the following references would be helpful [5–8].

Another thing is that we might be able to apply the method of mean curvature flow to generalize the isoperimetric inequality to higher-dimensional manifolds. For example here is a conjecture

Conjecture 5.1

Let M^n be a compact, boundaryless hypersurface embedded in n -dimensional Euclidean space, enclosing a bounded region Ω . Denoting the $(n+1)$ -dimensional volume of this region as $V(\Omega)$ and the n -dimensional surface area of M as $A(M)$. Then for some given positive real number A_0 there should be a hypersurface that maximizes $V(\Omega)$ under a fixed surface area $A(M) = A_0$. Furthermore, for any dimension n there should be a constant c_n that solely depends on n and also makes the following inequality holds.

$$\frac{A^{n+1}(M)}{V^n(\Omega)} \geq c_n \quad (5.1)$$

5.2 Review the Proof of the Multiplicity-One Conjecture and Try to Prove The Smale Conjecture

References

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A. Appendix of code and algorithm

A.1 code blocks

```

1  import torch
2  import torch.nn as nn
3  import torch.optim as optim
4  import numpy as np
5  import matplotlib.pyplot as plt
6  from mpl_toolkits.mplot3d import Axes3D
7  from scipy.integrate import odeint
8
9  """test"""
10 a=torch.tensor([[6,2,3],[3,4,5],[2,7,4],[5,2,0]])
11 b=torch.tensor([1,2,3,4])
12 b=b.unsqueeze(1)      # the text message
13 c=b*a
14 d=torch.tensor([1.0, 1.2, 1.5])
15 f=d+a
16 e1=f[:,1].unsqueeze(1)
17 Ar = np.array([[1,2,3],[4,5,6],[7,8,9]])
18 ar = torch.from_numpy(Ar)
19 print(Ar)
20 print(ar)

```

A.2 Algorithm

Algorithm 1: Simulation-optimization heuristic

Data: current period t , initial inventory I_{t-1} , initial capital B_{t-1} , demand samples

Result: Optimal order quantity Q_t^*

```

1   $r \leftarrow t$ ;
2   $\Delta B^* \leftarrow -\infty$ ;
3  while  $\Delta B \leq \Delta B^*$  and  $r \leq T$  do
4       $Q \leftarrow \arg \max_{Q \geq 0} \Delta B_{t,r}^Q(I_{t-1}, B_{t-1})$ ;
5       $\Delta B \leftarrow \Delta B_{t,r}^Q(I_{t-1}, B_{t-1}) / (r - t + 1)$ ;
6      if  $\Delta B \geq \Delta B^*$  then
7           $Q^* \leftarrow Q$ ;
8           $\Delta B^* \leftarrow \Delta B$ ;
9      end if
10      $r \leftarrow r + 1$ ;
11 end while

```
